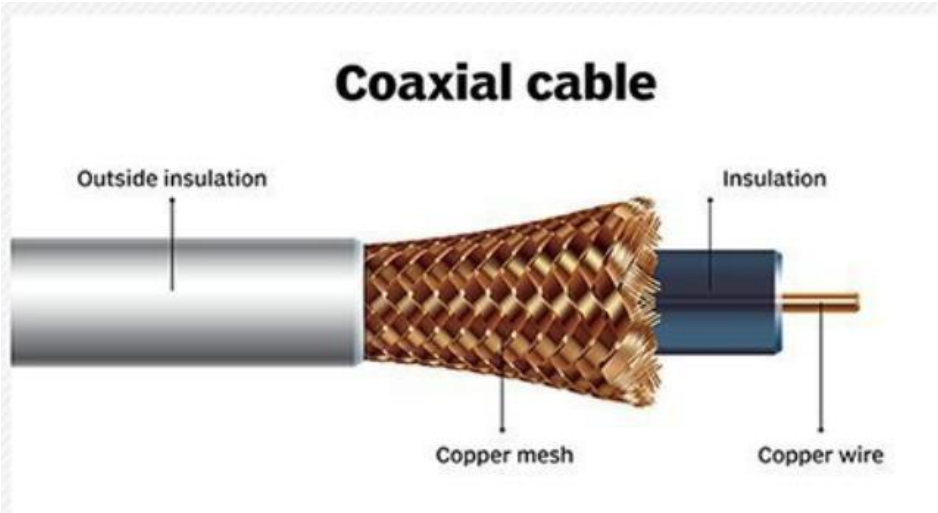
Practical 01

Practical 1: Identifying Network Component

a. **Aim:** Identify and document types of cables (Coaxial, Twisted Pair, Fiber Optic)

Coaxial Cable:

Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable. Ithas a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two. The outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.



Applications:

- Coaxial cable was widely used for both analog and digital data transmissions.
- It has higher bandwidth.
- Inexpensive when compared to fiber optical cables.
- It uses for longer distances at higher data rates.
- Excellent noise immunity.
- Used in LAN and Television distribution.

Disadvantage :

- Single cable failure can fail the entire network.
- Difficult to install and expensive when compared with twisted pairs.
- If the shield is imperfect, it can lead to grounded loop.

Twisted pair cable:

A twisted pair cable is a type of cable made by putting two separate insulated wires together in a twisted pattern and running them parallel to each other. This type of cable is widely used in different kinds of data and voice infrastructures.

Twisted pair is of two types:

1. Shielded Twisted Pair(STP)
2. Unshielded Twisted Pair(UTP)

Shielded Twisted Pair:

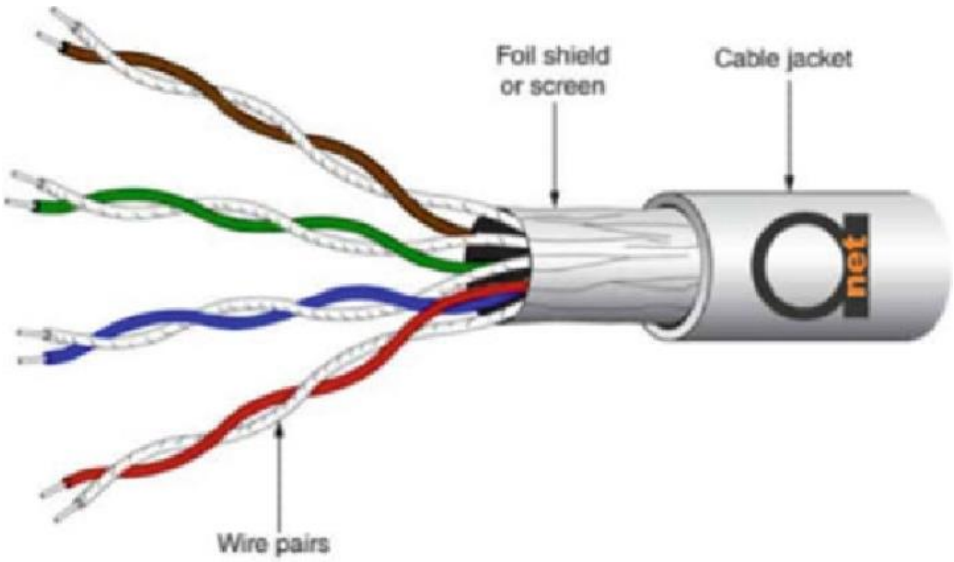
Shielded Twisted Pair (STP) cables additionally have an overall conducting metallic shields covering four twisted pair wires. There may be another conducting metallic shields covering individual twisted pairs also. These metallic shields blocks out electromagnetic interference to prevent unwanted noise from the communication circuit.

Advantage of Shielded Twisted Pair:

1. The cost of the shielded twisted pair cable is not very high and not very low.
2. An installation of STP is easy.
3. It has higher capacity as compared to unshielded twisted pair cable.
4. It has a higher attenuation
5. It is shielded that provides the higher data transmission rate.

Disadvantages:

1. It is more expensive as compared to UTP and coaxial cable.
2. It has a higher attenuation rate.



Unshielded Twisted Pair(UTP)

An unshielded twisted pair is widely used in telecommunication. It is most common type when compared with shielded twisted pair cable which consists of two conductors usually copper, each with Its own colour plastic insulator

Advantages Of Unshielded Twisted Pair:

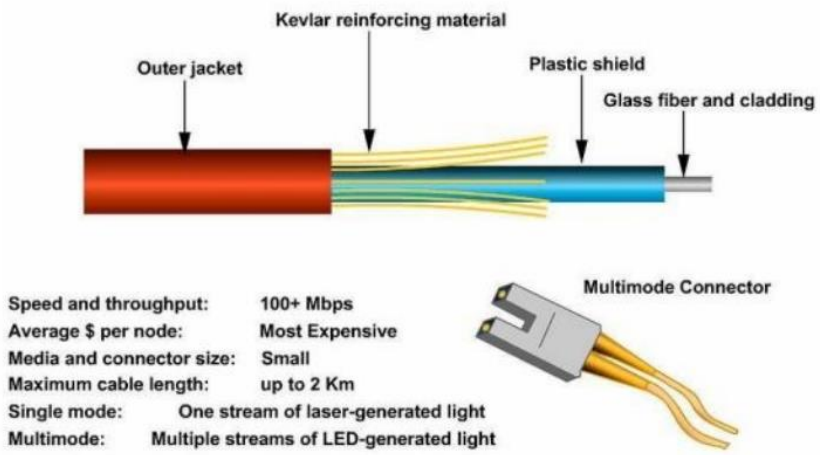
1. It is cheap.
2. Installation of the unshielded twisted pair is easy.
3. It can be used for high-speed LAN.

Disadvantage:

1. The cable can only be used for shorter distances

Fiber Optic Cable:

A fiber optic cable is a network cable that contains strands of glass fibers inside an insulated casing. They're designed for long-distance, high-performance data networking, and telecommunications. compared to wired cables, fiber optic cables provide higher bandwidth and transmit data over longer distances. Fiber optic cables support much of the world's internet, cable television, and telephone systems.



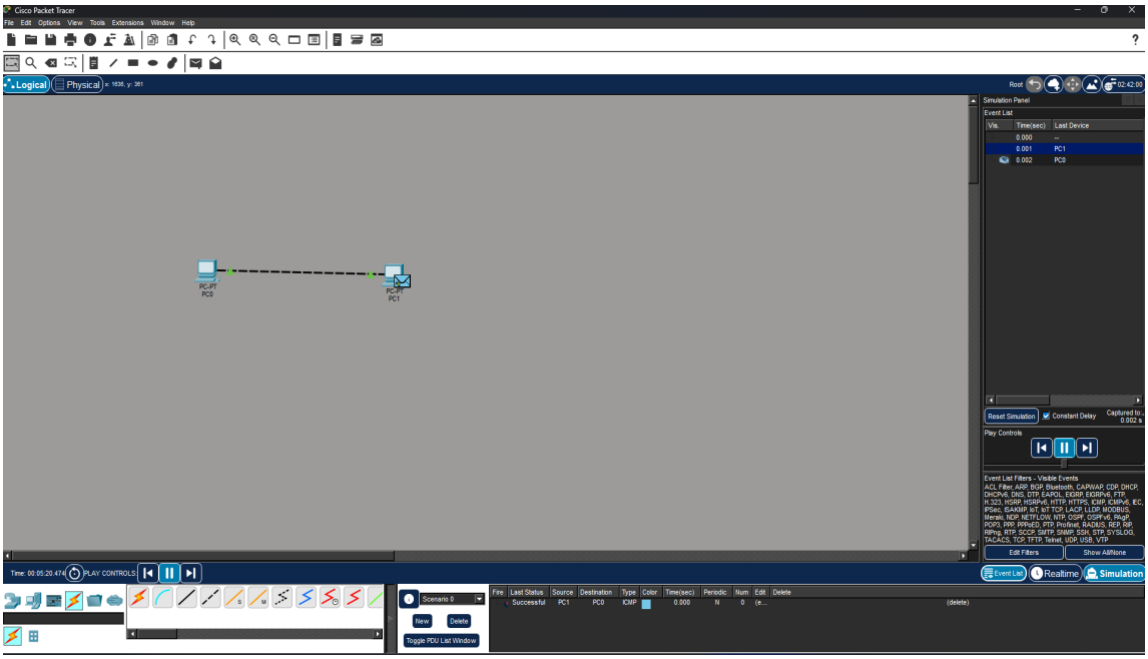
Advantages of Fiber Optic Cables:

1. The loss of signal in optical fiber is less than that in copper wire.
2. Opticalfibers usually have a longer life cycle for over 100 years.

Disadvantage:

1. It is expensive.
2. Difficult to install.

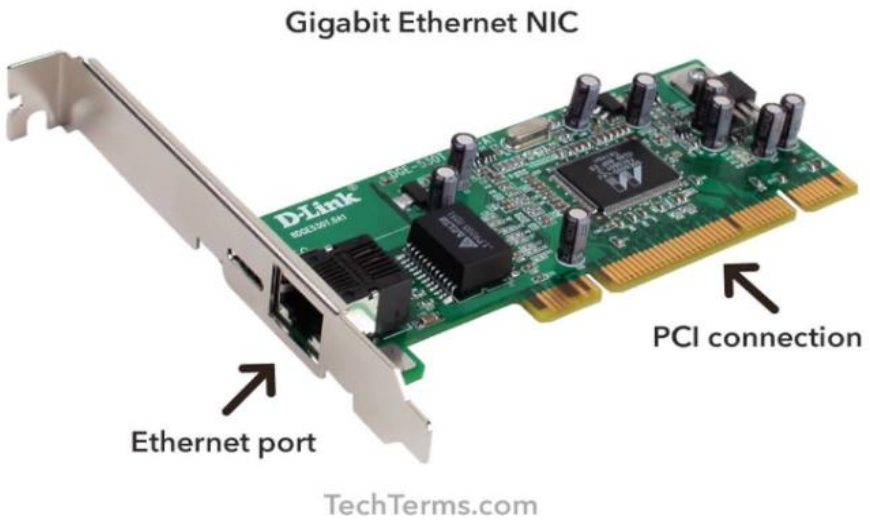
b. **AIM:** Demonstrate connecting PCs using a switch and straight-through cables.



c. **AIM:** Examine functions of NIC, hub, router, and switch physically or virtually.

Network Interface Card (NIC)

A hardware component inside a computer that connects it to a network.



Functions:

1. Provides a physical connection to the network (wired or wireless)
2. Assigns a unique MAC address to the device
3. Converts data into signals (electrical or wireless) for transmission
4. Receives incoming data and passes it to the system

Hub:

A basic networking device that connects multiple computers.

Functions:

1. Receives data from one device
2. Broadcasts the data to all connected devices
3. Works at the Physical Layer (Layer 1)

Limitations:

- No intelligence (cannot filter data)
- Causes network congestion

Switch:

A switch is a Networking device in a computer network that connects other devices together. Multiple data cables are plugged into a switch to enable communication between different networked Devices. switches manage the flow of data across a network by transmitting a received network packet Only to the one or more devices for which the packet is intended

Functions:

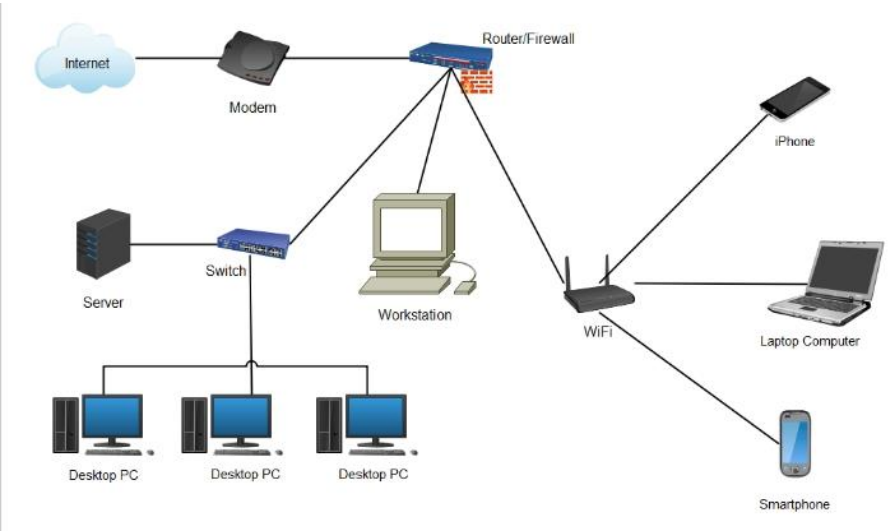
1. Sends data only to the intended device
2. Uses MAC addresses to identify devices
3. Reduces network traffic
4. Works at the Data Link Layer (Layer 2)

Advantages over hub:

- Faster and more efficient
- Avoids unnecessary data transmission

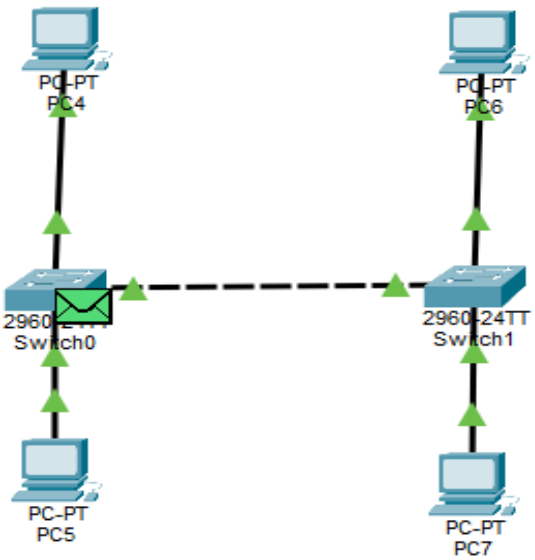
Router:

A router is an electronic device that interconnects two or more computer networks, and Selectively interchanges packets of data between them. Each data packet contains address Information that a router can use to determine if the source and destination are on the same network, Or if the data packet must be transferred from one network to another. Where multiple routers are Used in a large collection of interconnected networks, the routers exchange information about target System addresses, so that each router can build up a table showing the preferred paths between any two systems on the interconnected networks.

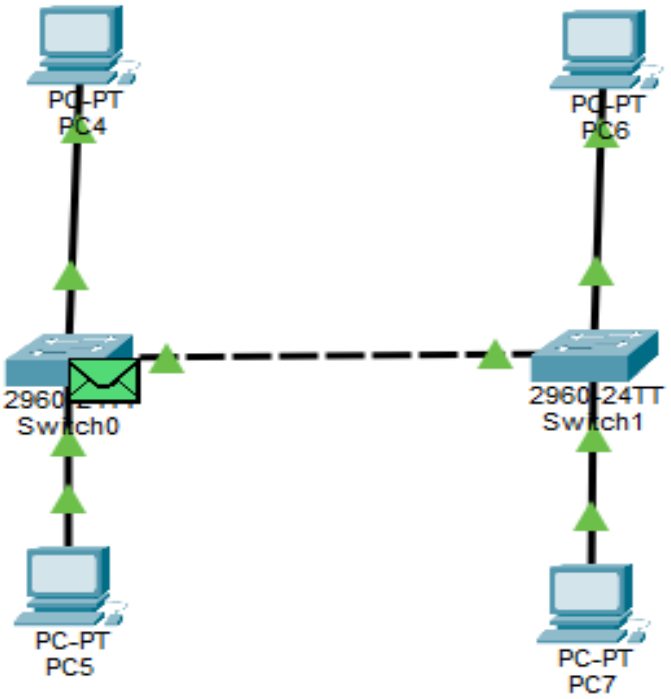


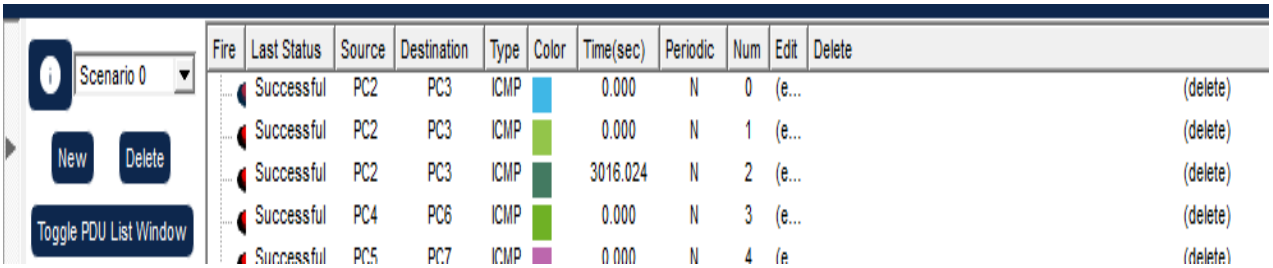
Functions:

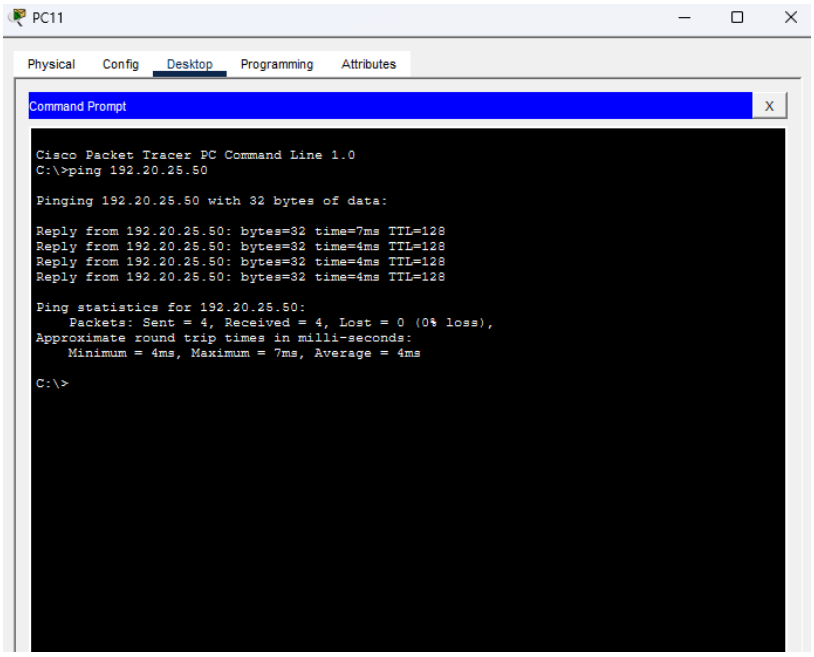
1. Routes data between networks using IP addresses
2. Connects LAN to WAN (Internet)
3. Provides Wi-Fi connectivity
4. Performs NAT (Network Address Translation)



d. AIM: Colour code for crimping LAN (Cat 5/6/7) cable



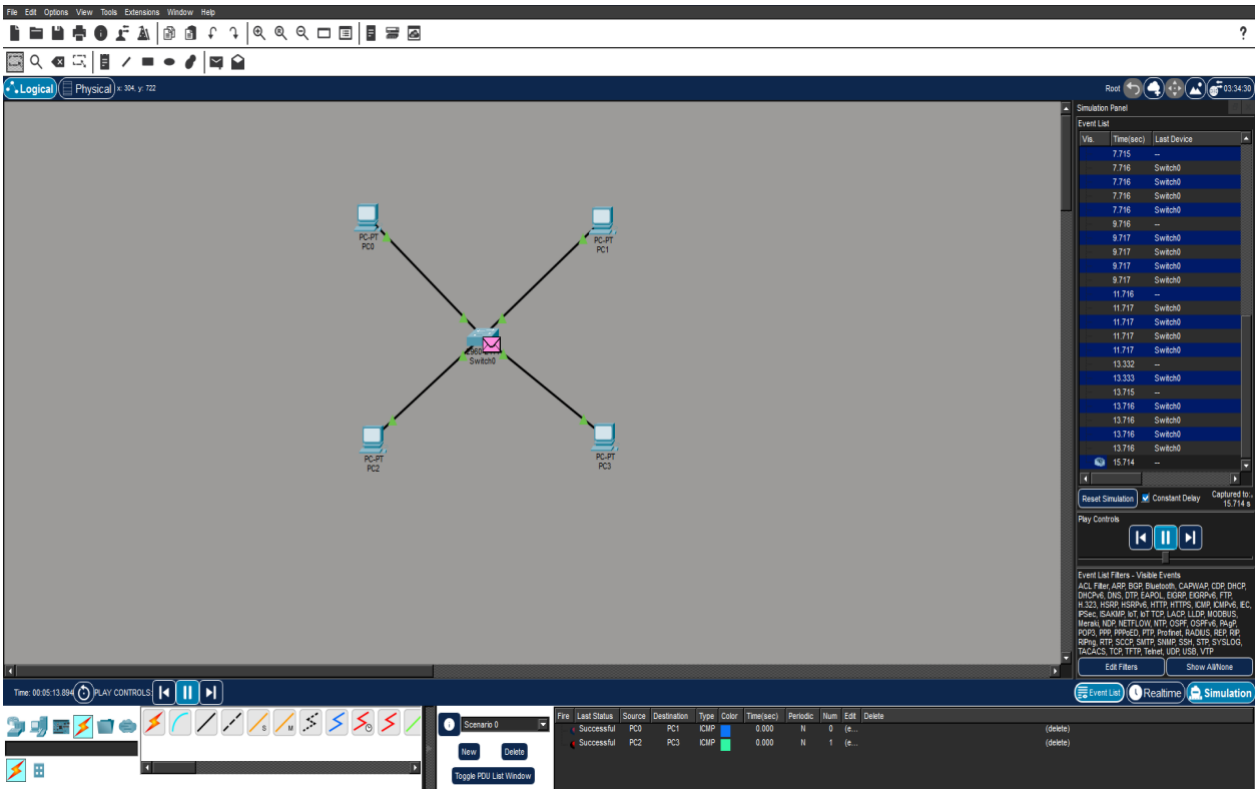




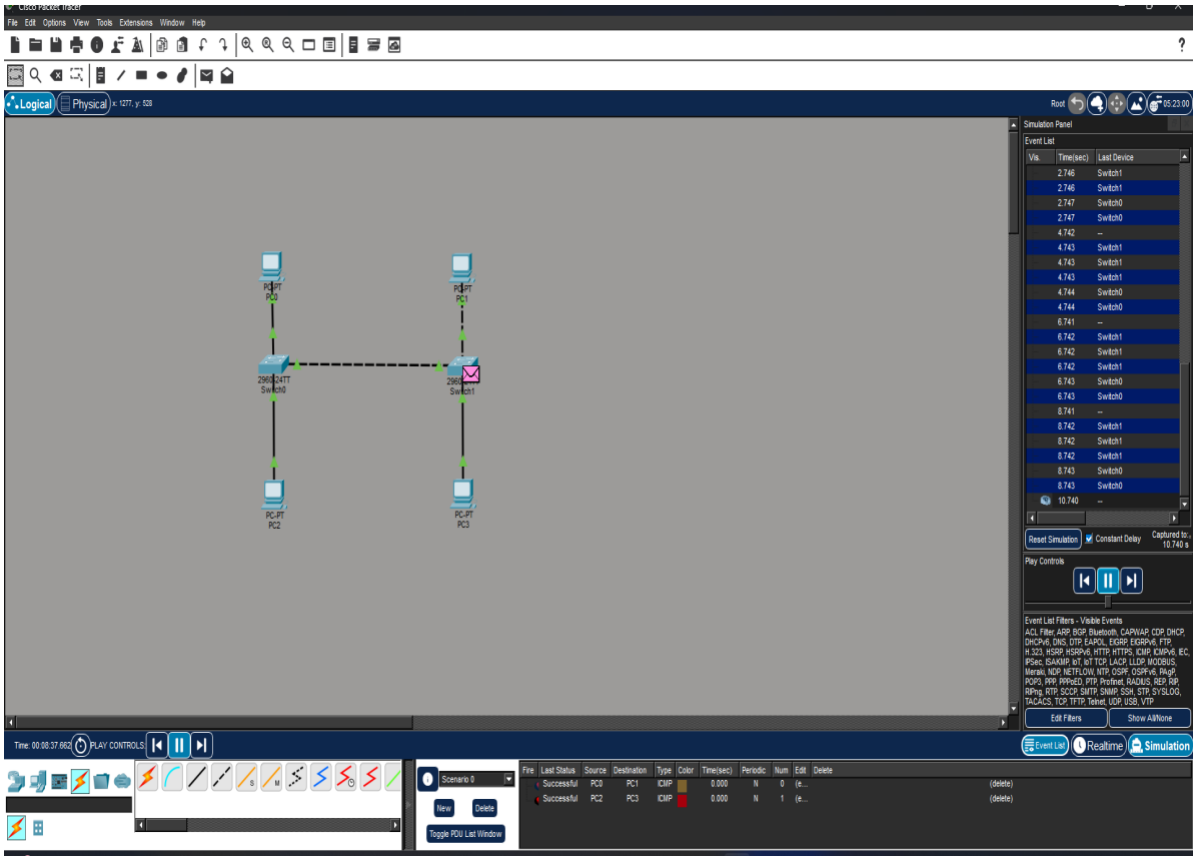
Practical 02

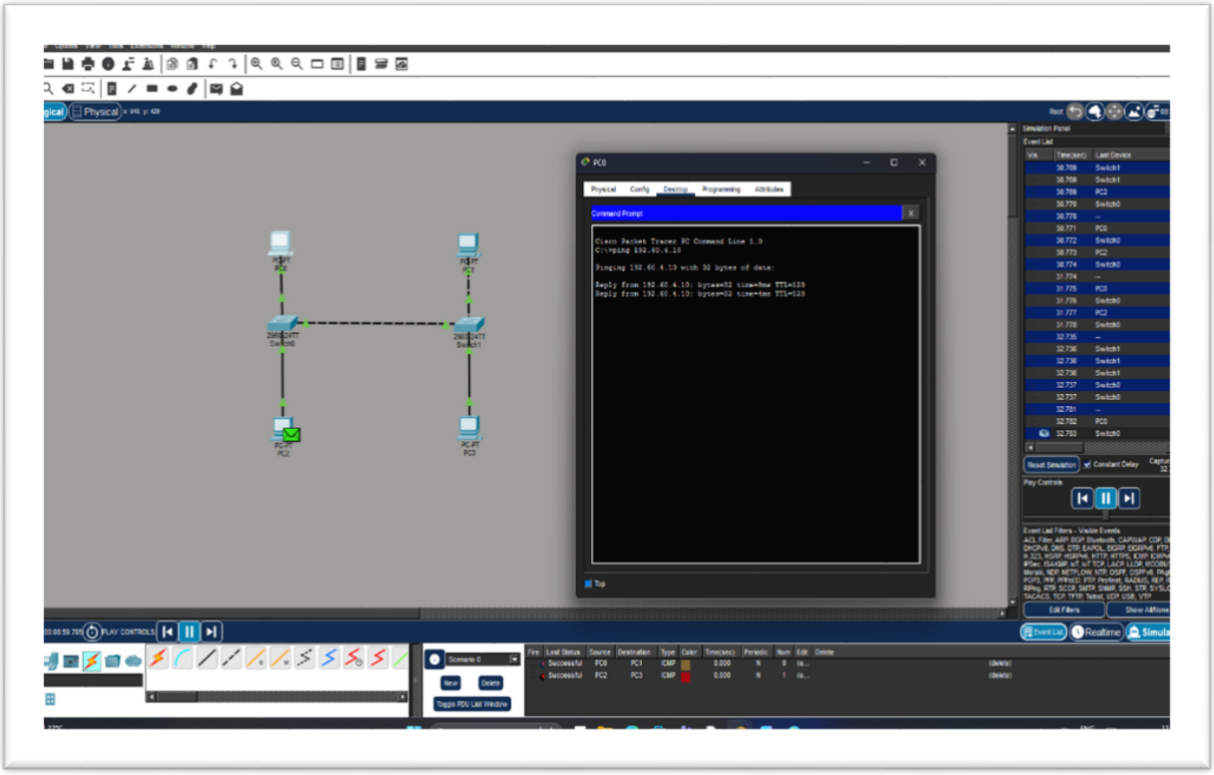
Practical 2: Understanding Network Topologies

a. **AIM:** Create a star topology in Packet Tracer.

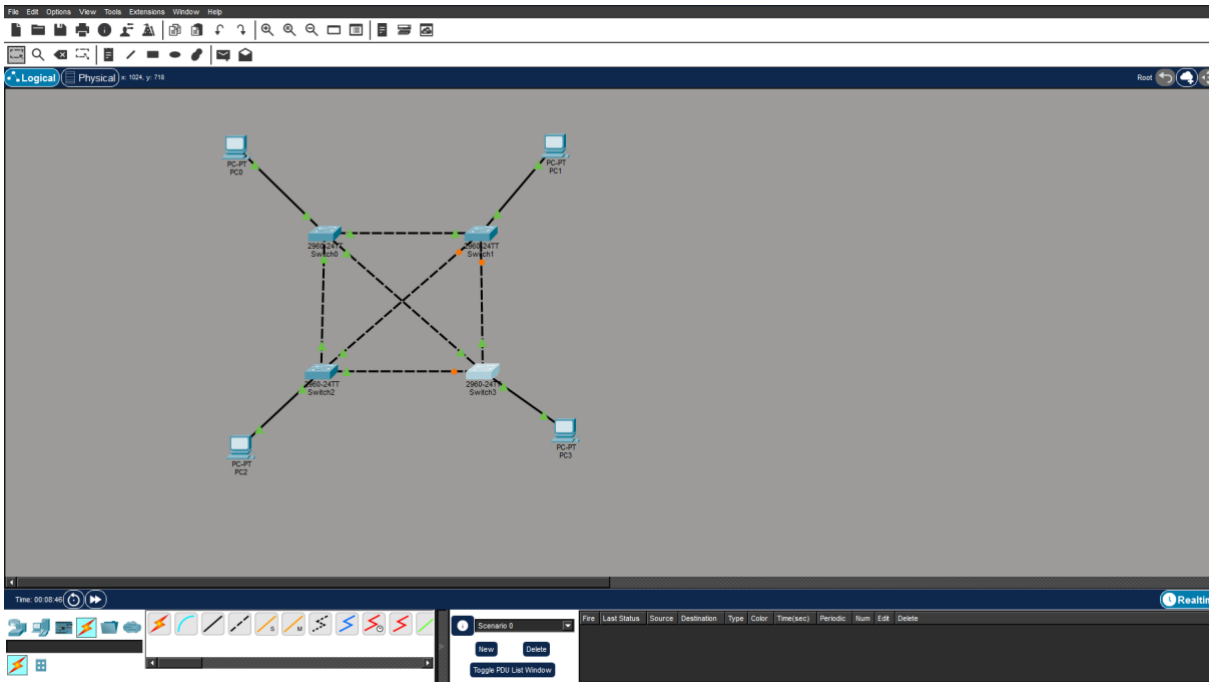


b. AIM: Simulate a bus topology and analyse traffic flow.

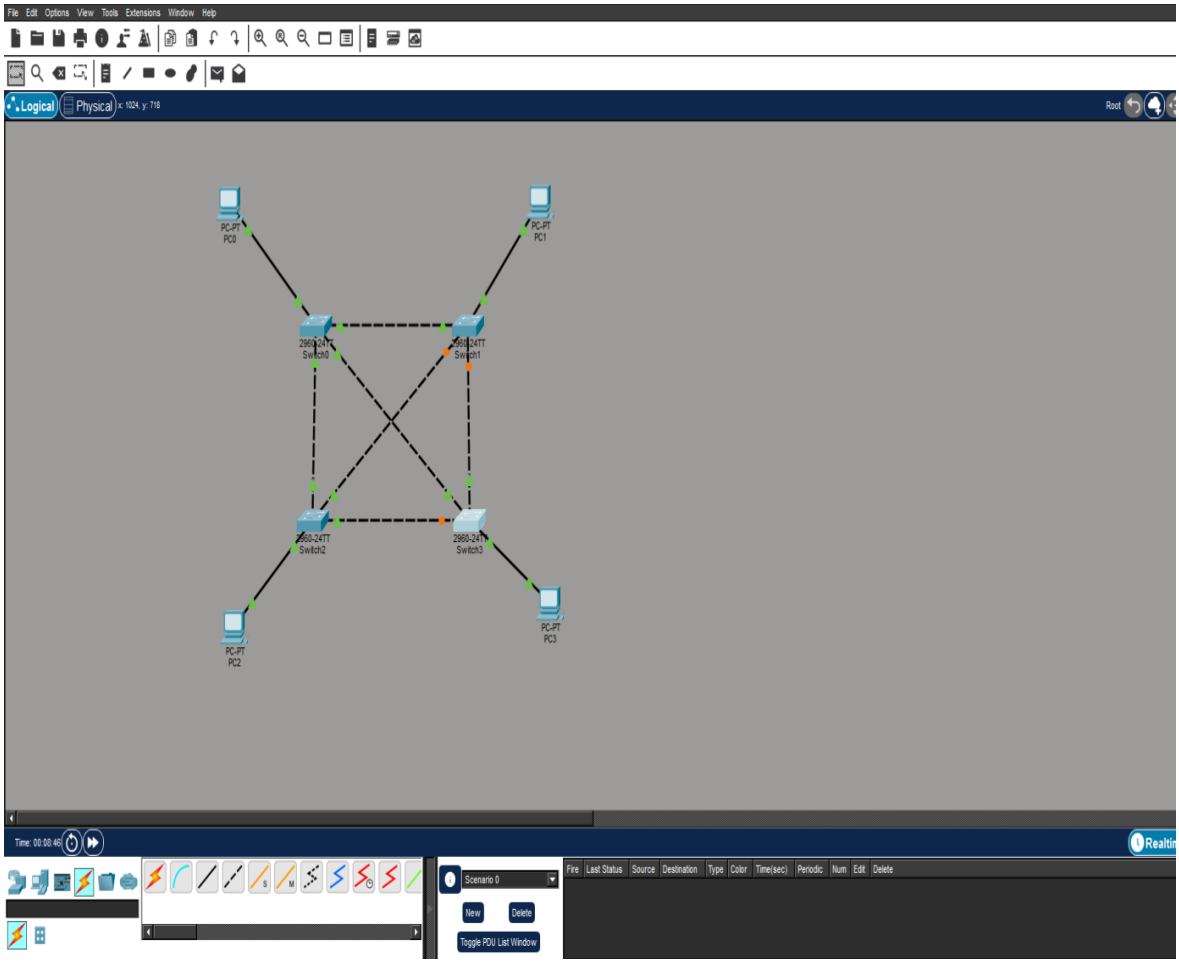




c. **AIM**: Demonstrate data communication in mesh topology.



d. AIM: Study of basic network command and Network configuration commands. Ipconfig, netstat, ARP etc.




```
acer PC Command Line 1.0
fig /all

 Connection:(default port)

tion-specific DNS Suffix..:
al Address.................: 0060.5CBD.C726
ocal IPv6 Address.........: ::
ress.......................: 0.0.0.0
 Mask......................: 0.0.0.0
t Gateway..................: 0.0.0.0
rvers......................: 0.0.0.0
ervers.....................: 0.0.0.0
 Client DUID...............: 00-01-00-01-0C-87-DA-DB-00-01-43-3B-BB-BB

net0 Connection:

tion-specific DNS Suffix..:
al Address.................: 0001.433B.BBBB
ocal IPv6 Address.........: FE80::201:43FF:FE3B:7596
ress.......................: 10.1.1.12
 Mask......................: 255.0.0.0
t Gateway..................: 0.0.0.0
rvers......................: 0.0.0.0
```

Practical 03

Practical 3: Error Detection and Correction Techniques

a. **AIM:** Simulate a data transmission scenario using parity bit error detection.

Q. Consider data transmitted by the sender is E, A, C, q the received bites by the receiver are 10001011, 10000010, 11100111, 11100000 each character data has been applied by a simple parity check detect using simple parity check detection method whether the recevied data has any error or not if you which is the character error.

Solution :-

E = 69 = 1000101
A = 65 = 1000001
C = 67 = 1000011
q = 113 = 1110001

Sender side

Data	Binary Data	Parity	Data to be sent
E	1000101	1	10001011
A	1000001	0	10000010
C	1000011	1	10000111
q	1110001	0	11100010

Receiver side

Data	Send data	Send parity	Received parity	error
10001011	10001011	1	1	Yes
10000010	10000000	0	0	Yes
11100111	10000111	1	1	Yes
11100000	11100010	0	1	No

error

b. **AIM:** Implement a CRC (Cyclic Redundancy Check) calculator and simulate bits errors.

CRC :-

Cyclic Redancy check.

- It is based on binary division.
- Total bits = message + Redantant bits
- It can find single bit burst, errors = polynominal length.
- It is the most commonly used error detection method.

Q. Data to be sent is 1010101010 and the polynomial redundancy bits are $x^4 + x^3 + 1$

A] Find the message sent by the sender.
B] Padded bits
C] If the data received by the receiver is 1010101010100100 has error or not.
D] Length of message.

Solution :-

$x^4 + x^3 + 1 = 1x^4 + 1x^3 + 0x^2 + 0x^1 + 1$

$= 11001 \longrightarrow$ divisor bits

$10 + 4 = 14 \longrightarrow$ size

polynominal highest degree.

Division = module 2 $\longrightarrow$ XOR

```
11001 | 1010101010000
        11001
        011000
         11001
        00001101 0
            11001
            00011000
              11001
              00001 0
```

101010101000 10

```
11001 | 10101010100100001 0
        11001
        011000
         11001
        00001101 0
            11001
            00011010
              11001
              00011000
                11001
                0000110
```

c. **AIM:** Demonstrate Simple check and 2D check mechanism.

Simple Parity check :-

It is a base error detection technique in which extra bit falled parity bit is added to the data can take only bit error threre are 2 types of parity even parityhe parity is said so that the ones in the data ~~d~~ to even add parity the parity is said so that the number of one's in the data is add.

1	0	1	1	0	1	0	1

Using simple parity check

data	+	parity bit

If is in data is add parity added is '1'
If is in data is even parity added is '0'

In given ex –

data

1	0	1	1	0	1	0	1	1

parity

Data Set

1	0	1	1	0	1	0	1	1

Q. Sender data E, A, C, q
Received data 10001011, 10000010, 11100111, 11100000
finds the error using 2D parity check.

Solution :-

Sender side

Data								Column parity
E	1	0	0	0	1	0	1	1
A	1	0	0	0	0	0	1	0
C	1	0	0	0	0	1	1	1
q	1	1	1	0	0	0	1	0
Row parity	0	1	1	0	1	1	0	

							Co	
1	0	0	0	1	0	1	1	1
1	0	0	0	0	0	1	0	0
1	1	1	0	0	1	1	1	1
1	1	1	0	0	0	0	1	0
0	0	0	0	1	1	1		
0	1	1	0	1	1	0		
✓	✗	✗	✓	✓	✓	✗		

Practical 04

Practical 4: IP Addressing and Subnetting

a. **AIM:** Convert between binary and decimal IPs

IP 205.16.37.39/27

27	5
block / Network	host

No of host - $2^5 = 32$

Network mask
11111111.11111111.11111111.11100000
255. 255. 255. 224

1st IP of the block:
255.255.255.224/27
AND
205.16.37.39/27

1st IP; 205.16.37.32/27

Last IP of the block:
205.16.37.63/27

Note: for the last IP consider host bits
for the first IP consider network bits

b. **AIM:** Subnet a class C IP address and assign subnets

IP 216.21.5.0

Sol: class C

Default mask
11111111.11111111.11111111.00000000

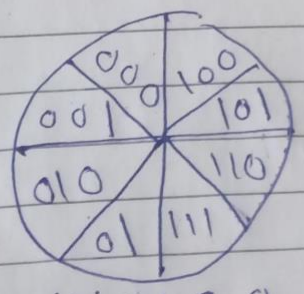
No of host = 30 = 1110 = 5 bits

subnet generator = 8 − 5 = 3

octet position = 255.255.255.11100000

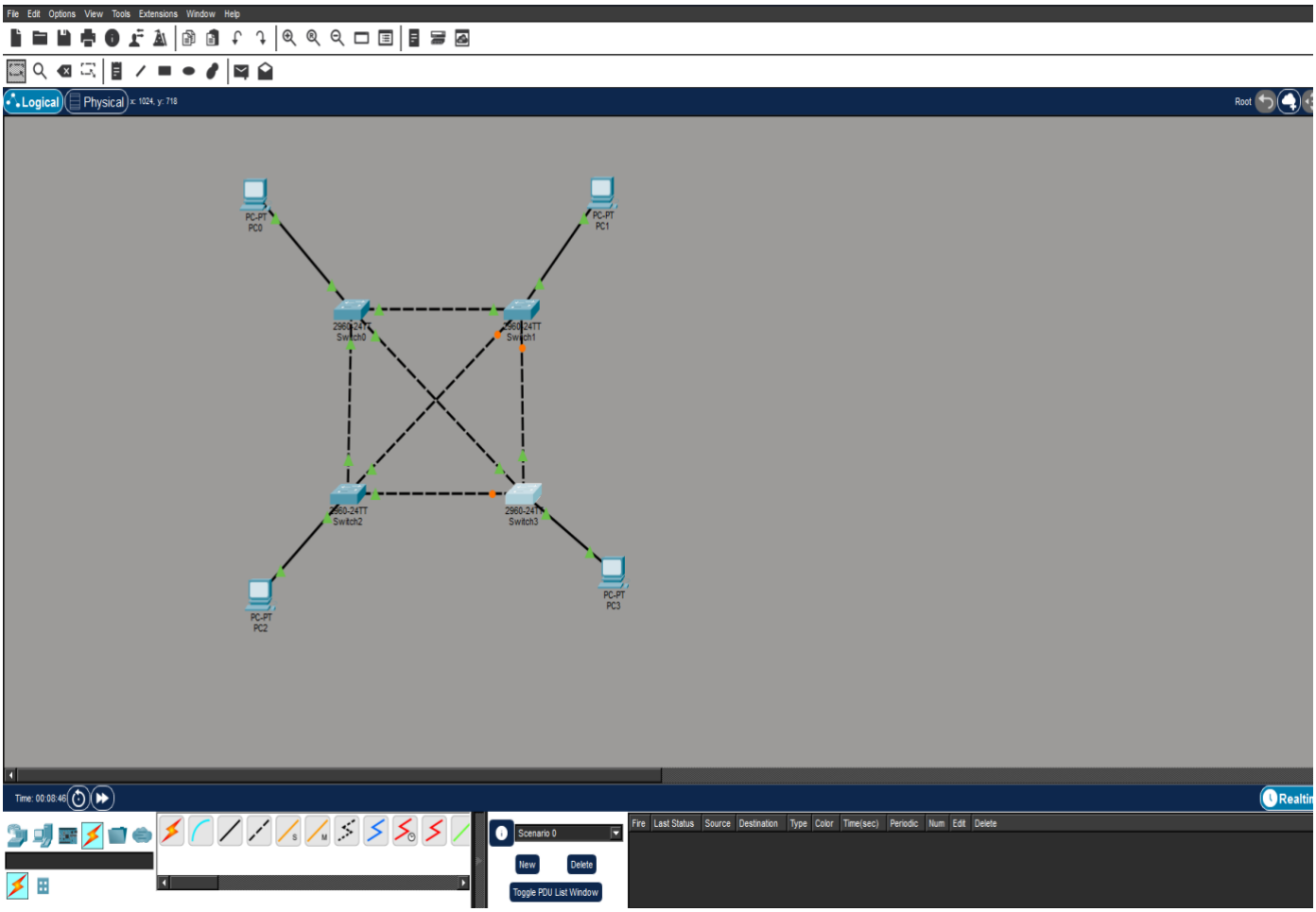
Network id : 216.21.5.0
Broadcast id : 216.21.5.254

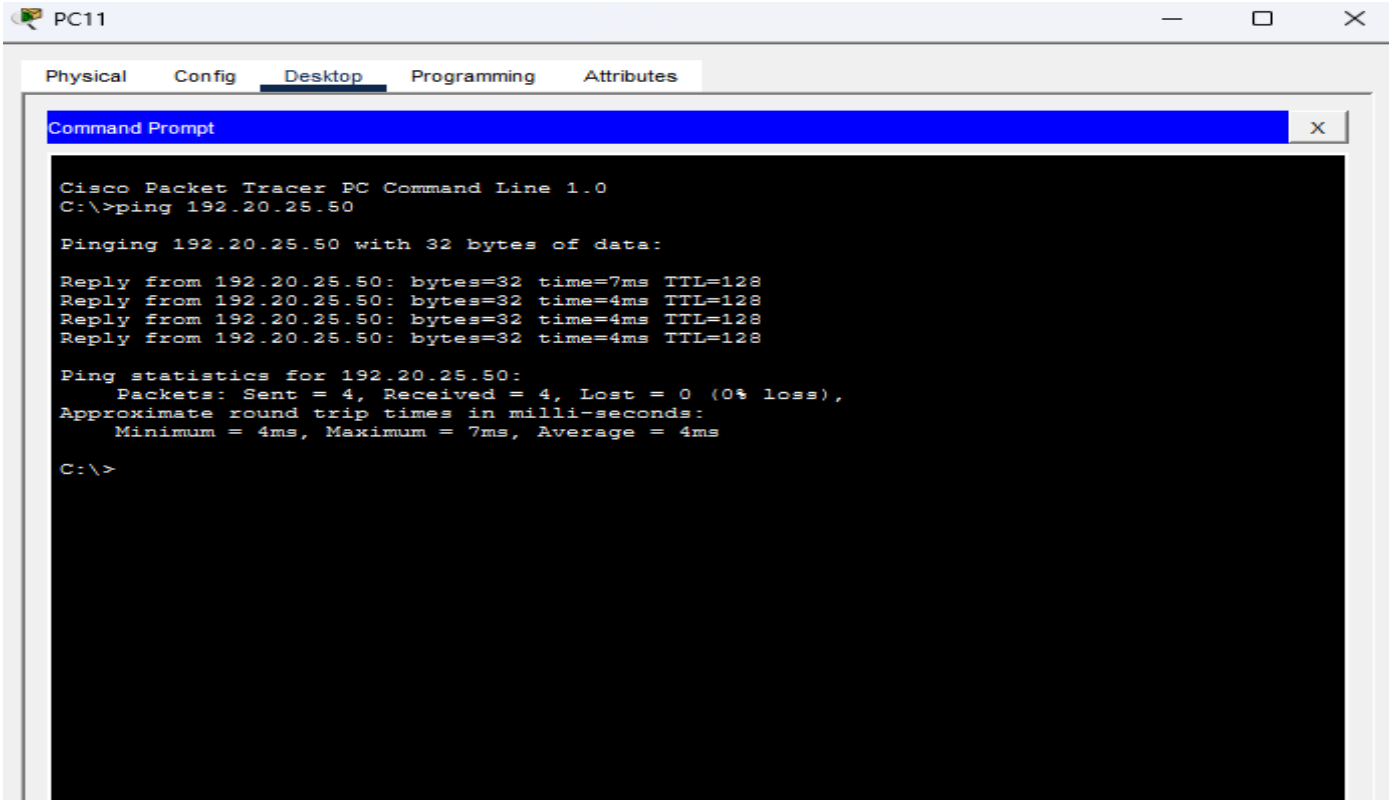
subnet group 1 : 216.21.5.0 to 216.21.5.31
subnet group 2 : 216.21.5.32 to 216.21.5.63
subnet group 3 : 216.21.5.64 to 216.21.5.95
subnet group 4 : 216.21.5.96 to 216.21.5.127
subnet group 5 : 216.21.5.128 to 216.21.5.159
subnet group 6 : 216.21.5.160 to 216.21.5.191
subnet group 7 : 216.21.5.192 to 216.21.5.223
subnet group 8 : 216.21.5.224 to 216.21.5.254



216.21.5.0

c. **AIM:** Configure IPs on Packet Tracer devices.





Router0

Physical Config CLI Attributes

IOS Command Line Interface

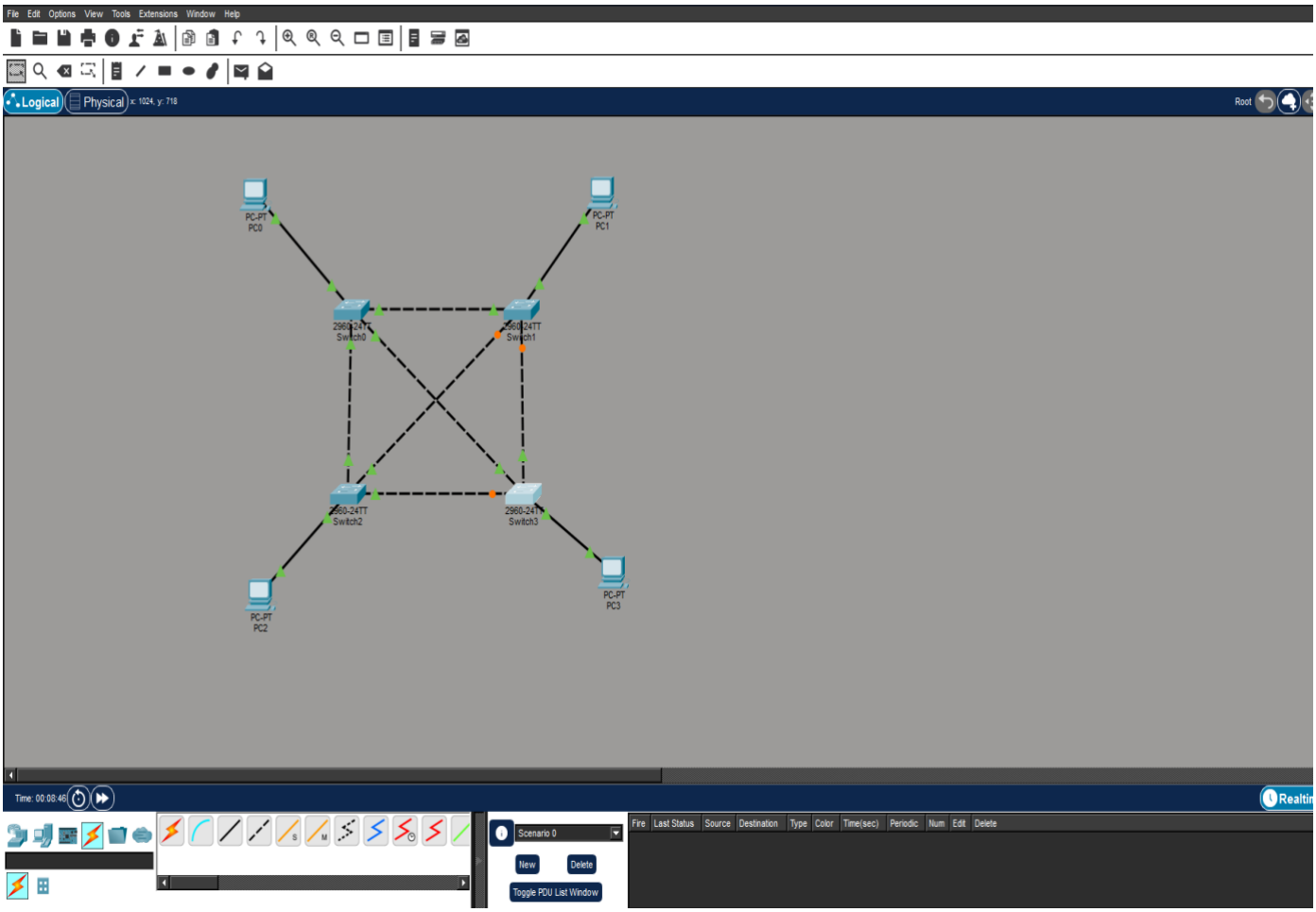
```
Router>en
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fa0/1
Router(config-if)#ip address 192.168.10.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.20.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

Router(config-if)#exit
Router(config)#interface se0/0/0
Router(config-if)#ip address 192.168.50.1 255.255.255.0
Router(config-if)#no shutdown
```

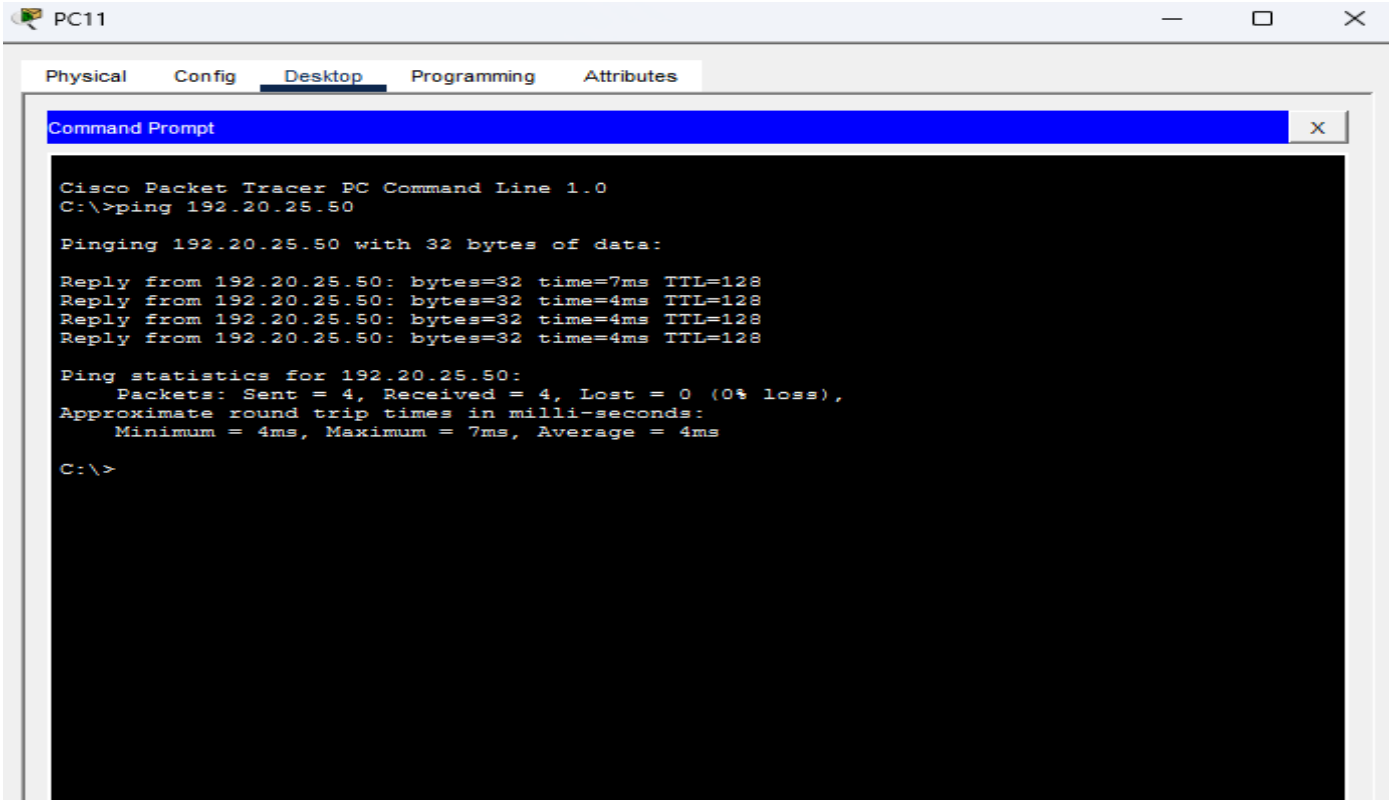
Ctrl+F6 to exit CLI focus

Copy Paste

Top

d. **AIM:** Validate connectivity using ping and trace route.

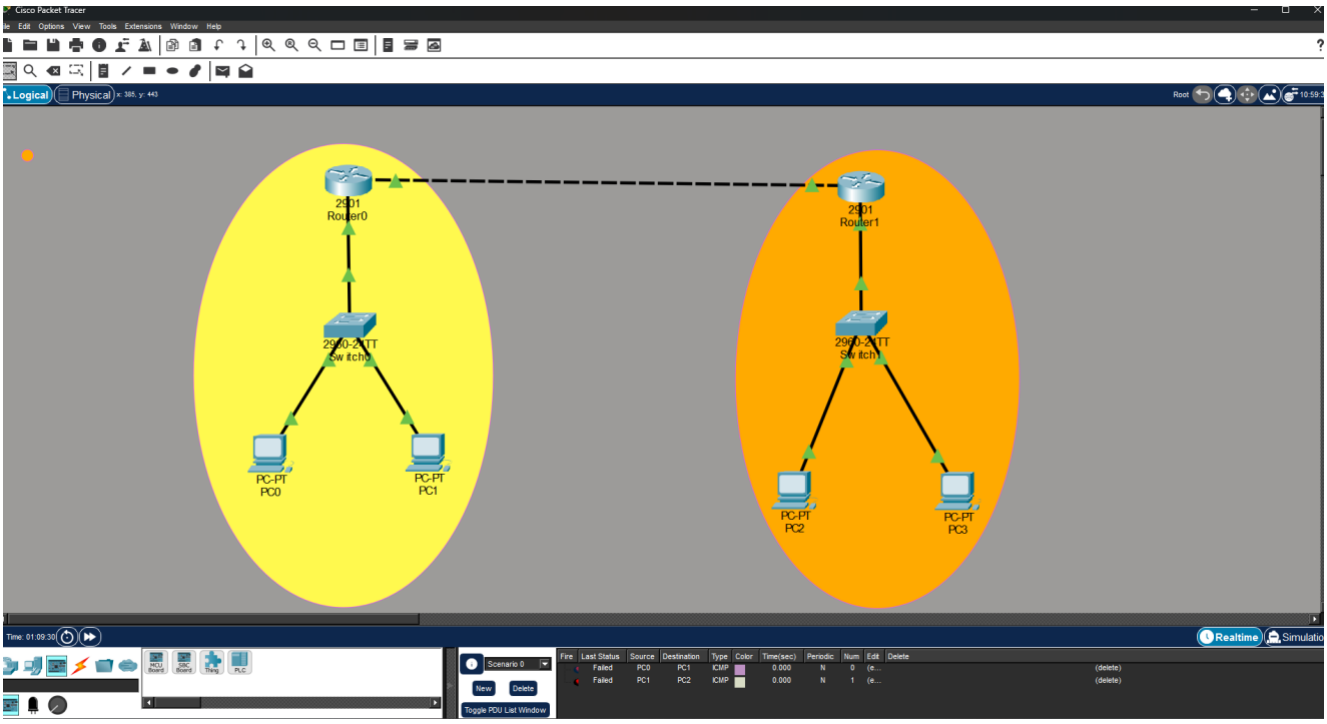


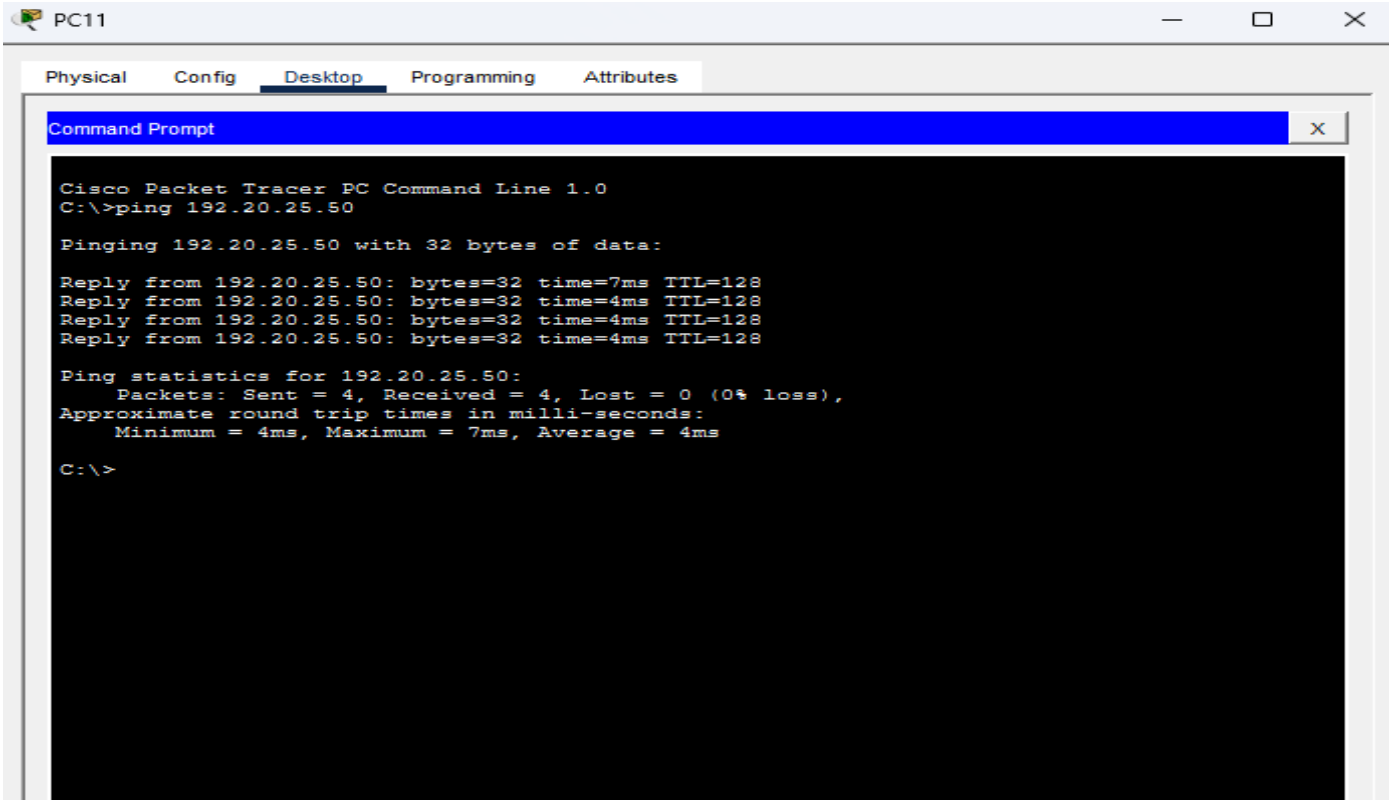


Practical 05

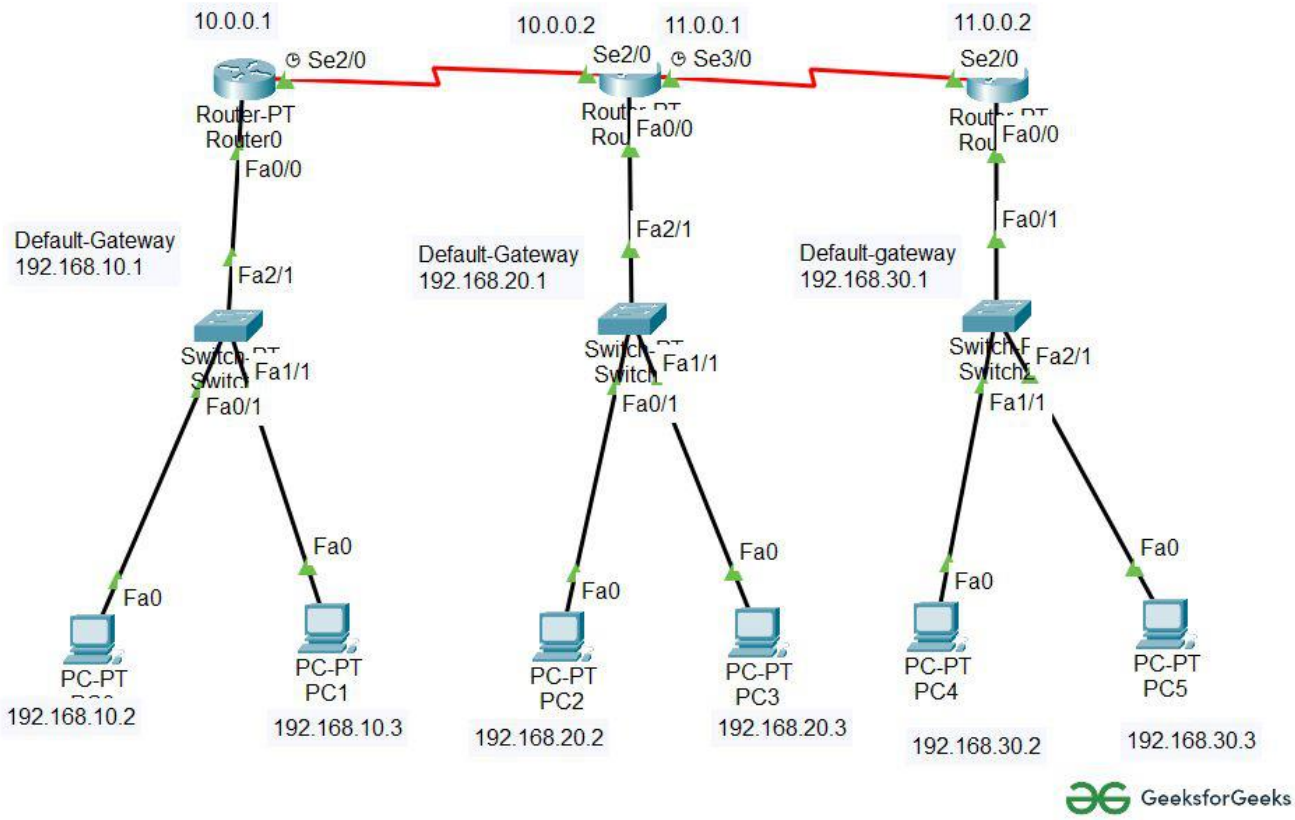
Practical 5: Routing Protocols in Action

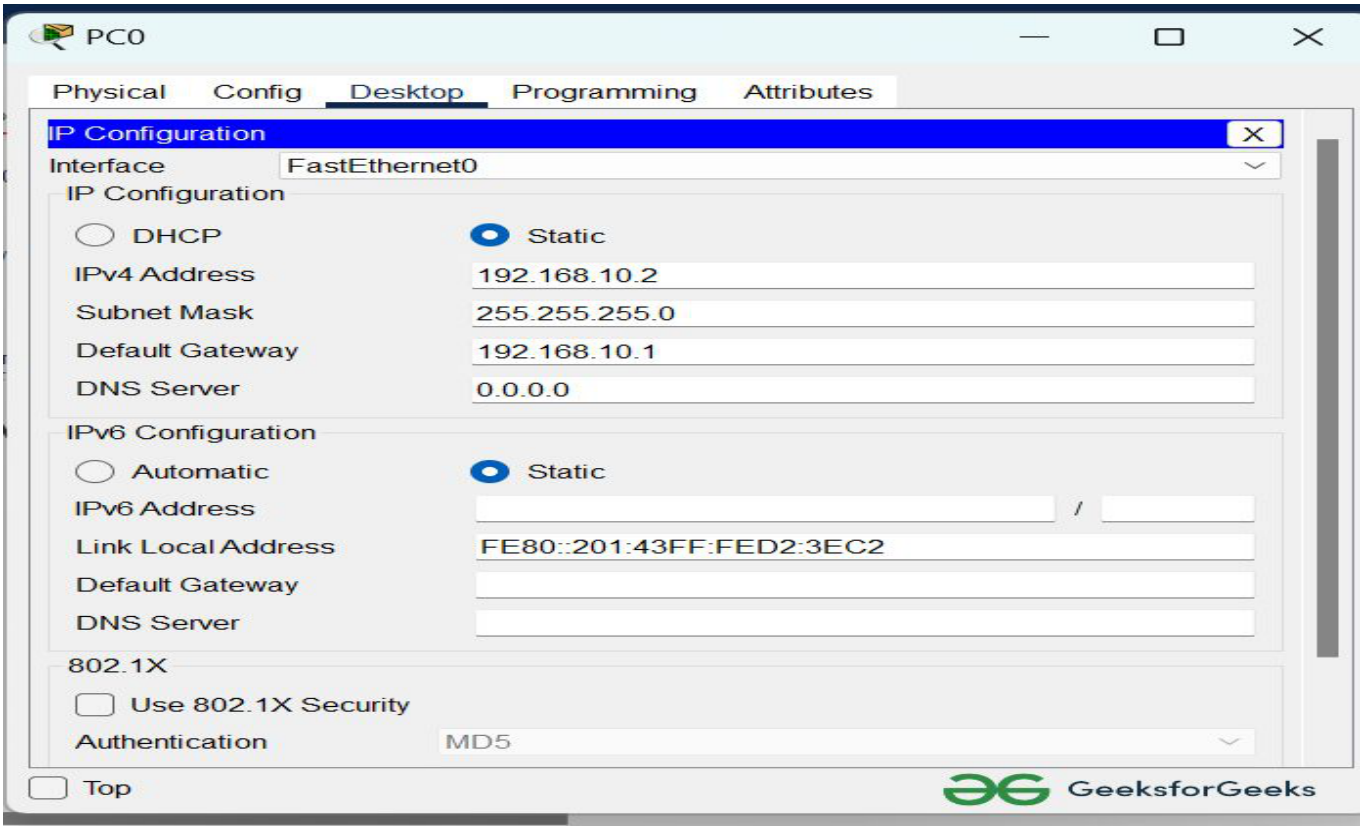
a. **AIM:** Implement static routing between 2 routers.



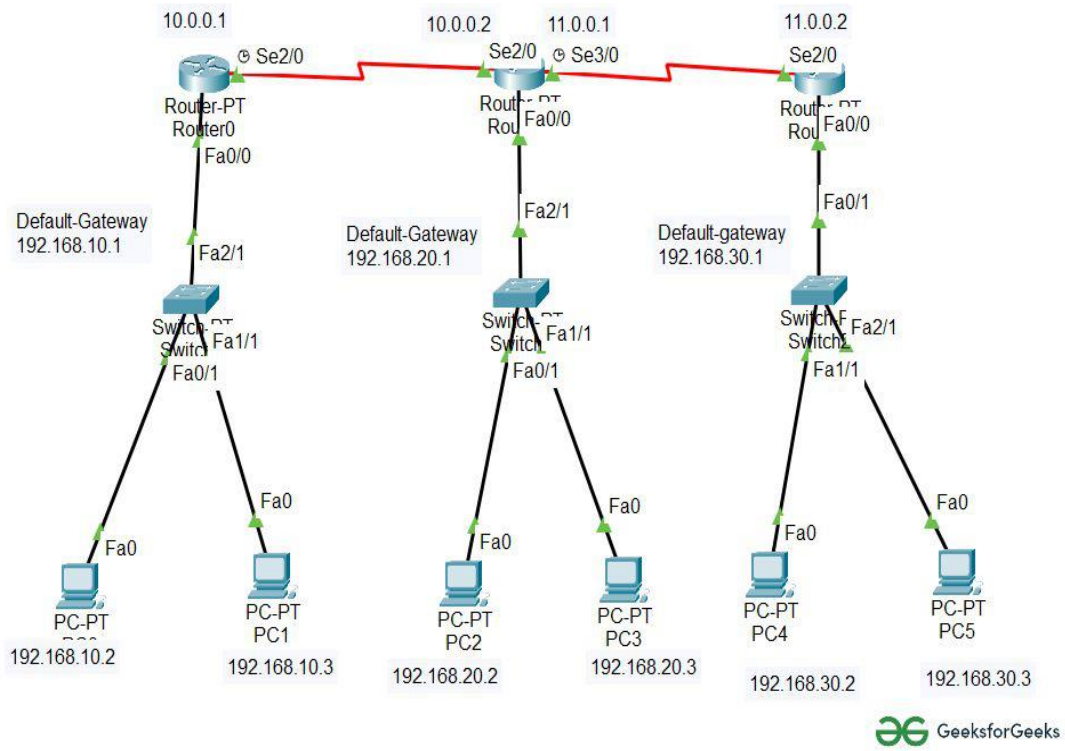


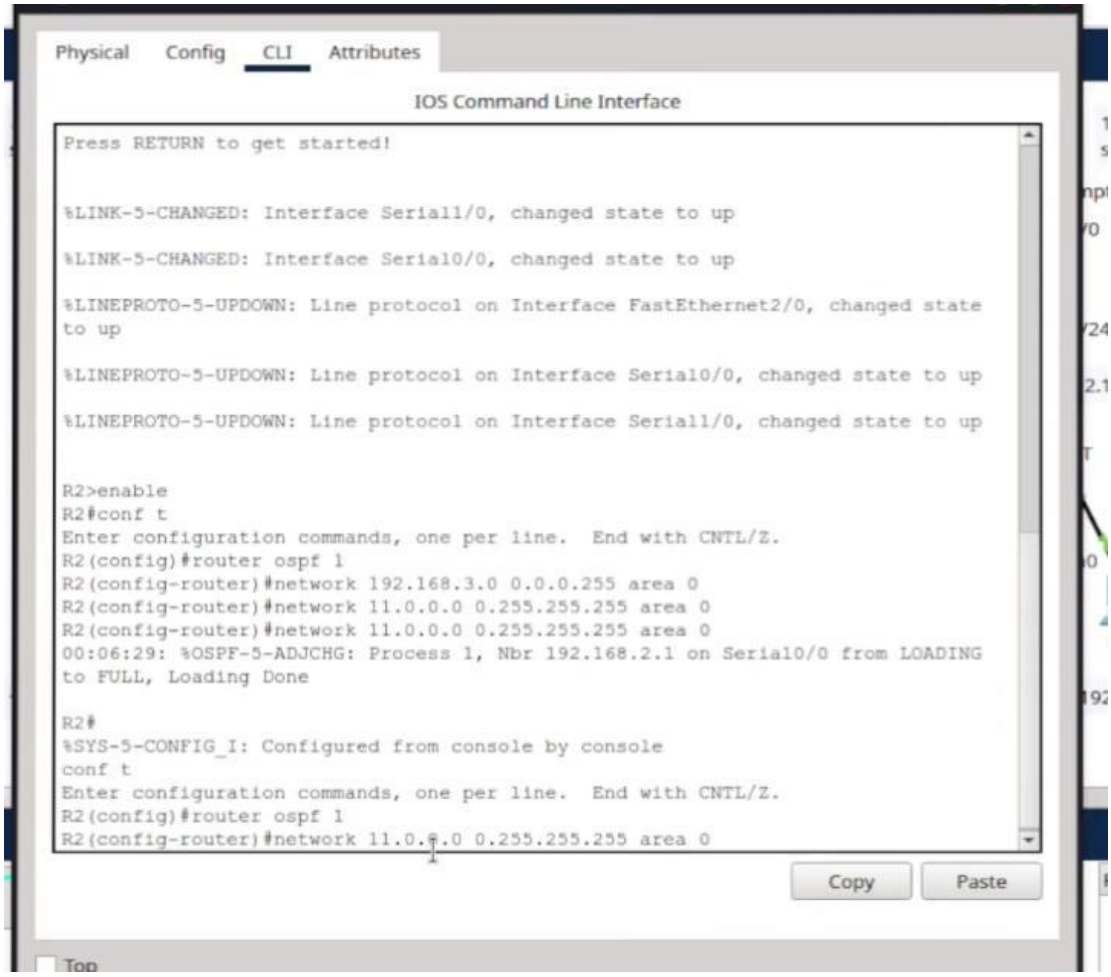
b. **AIM:** Configure RIP on a three-router topology.



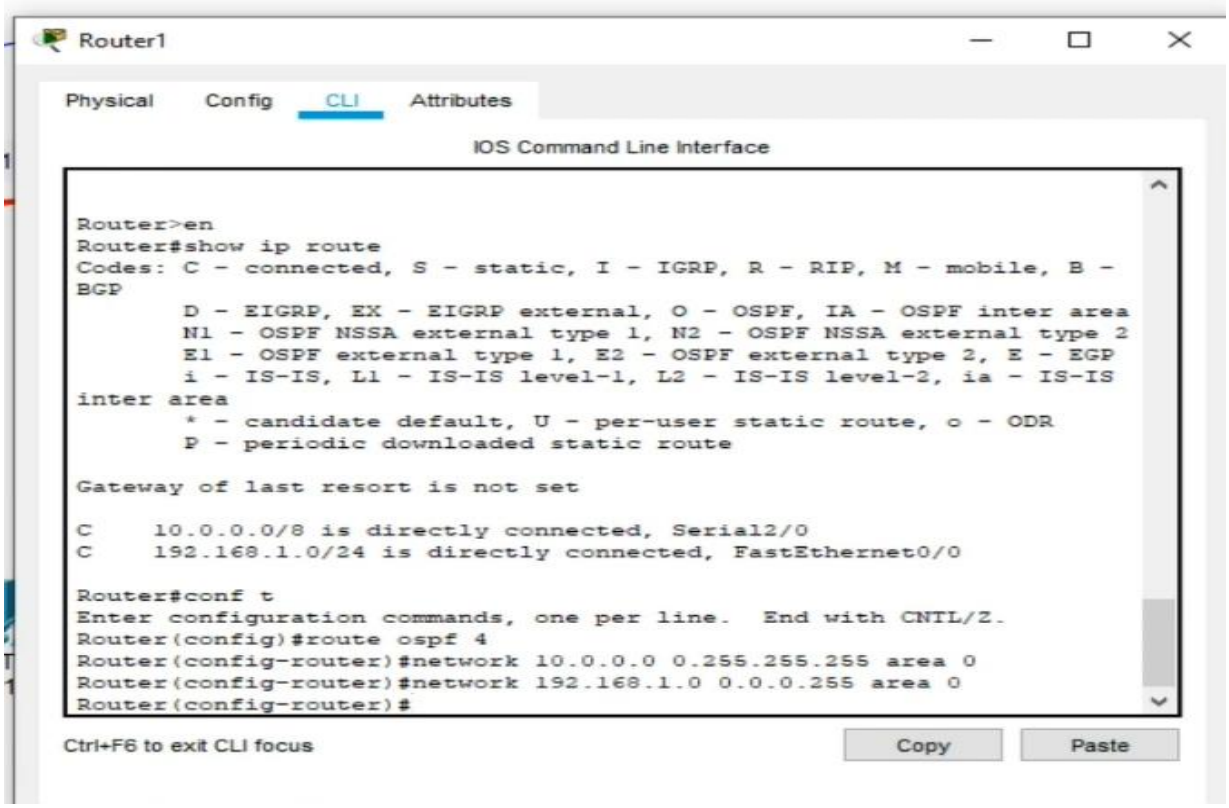


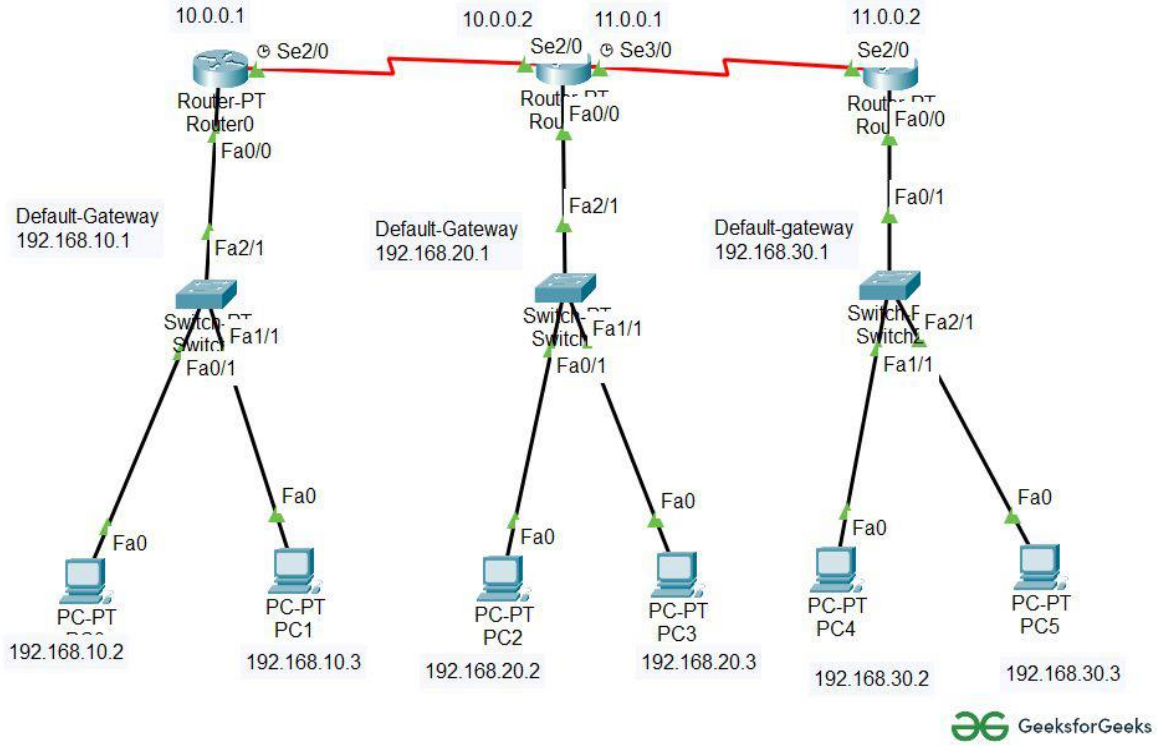
c. **AIM:** Set up simple OSPF with area configuration.





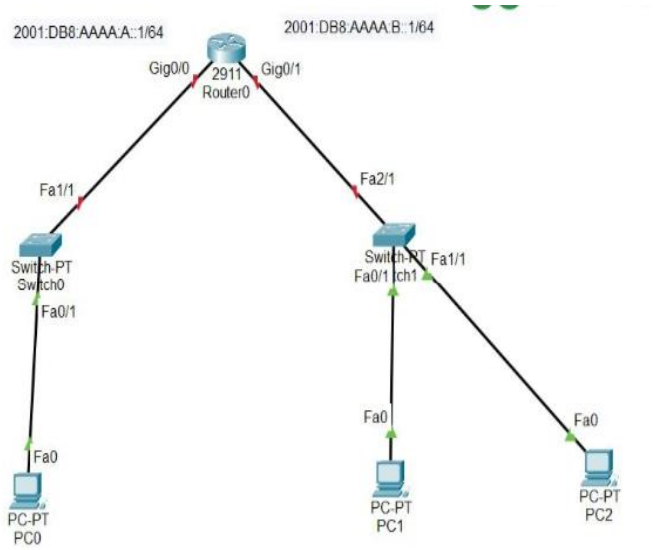
d. **AIM:** Set up multiarea OSPF with area configuration.

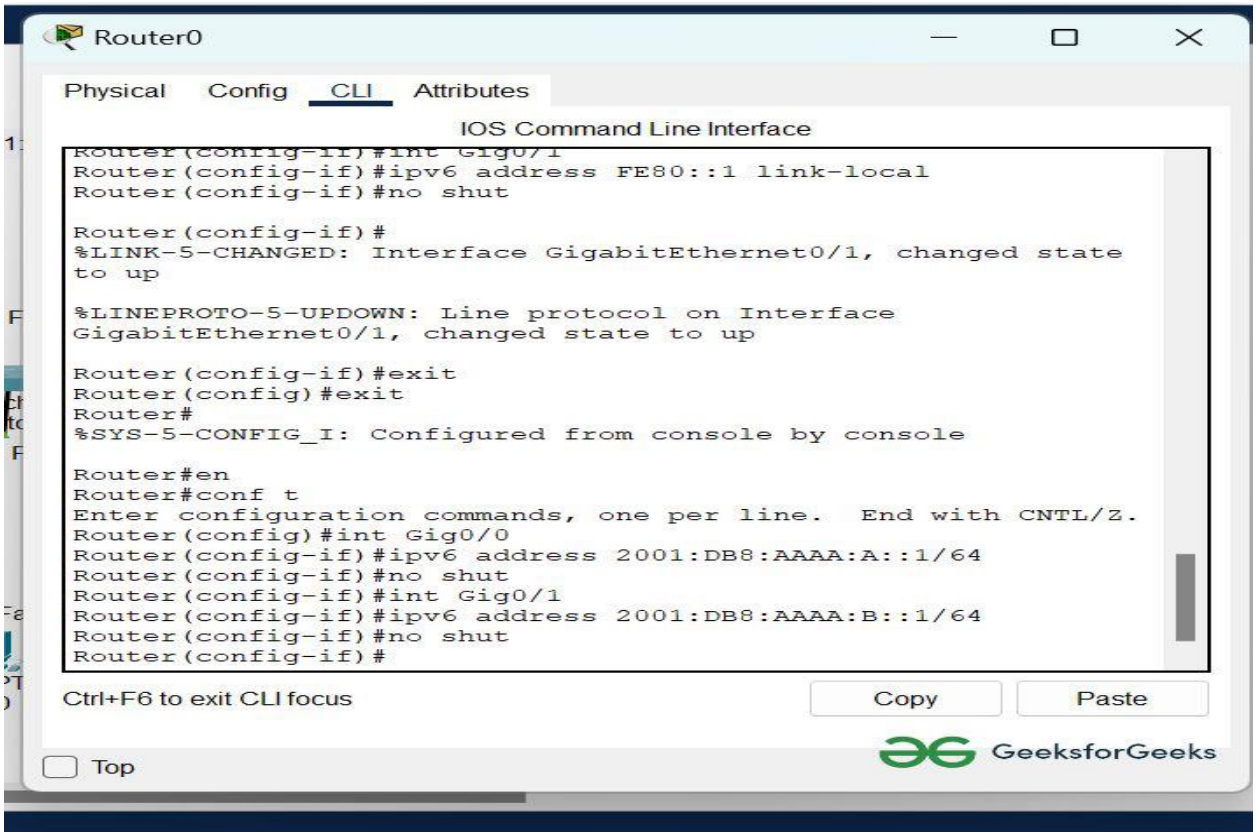




Practical 6: IPv6 and ICMP Tools

a. **AIM:** Configure IPv6 addresses in Packet Tracer.





PC1

Physical Config Desktop Programming Attributes

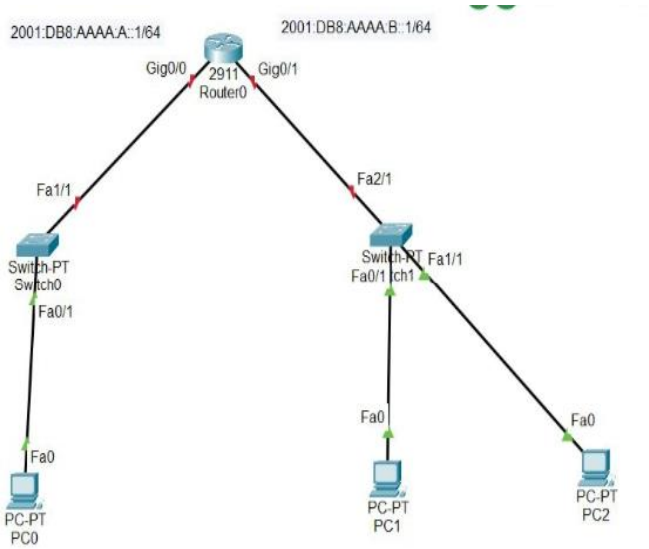
Command Prompt

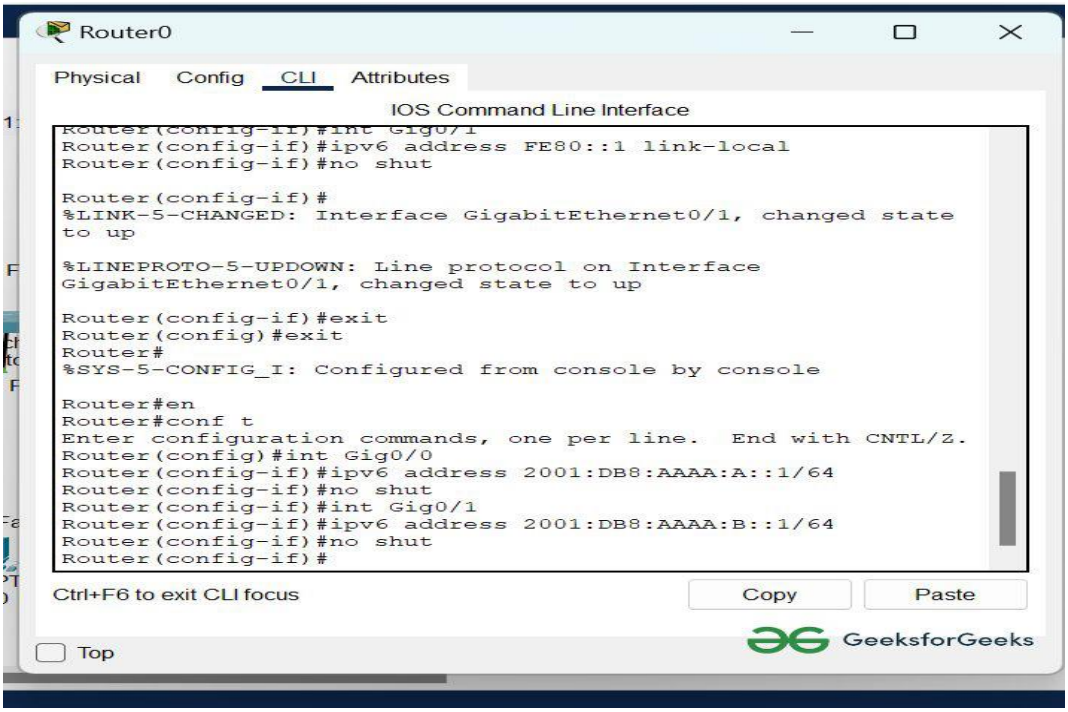
```
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:DB8:AAAA:A:20D:BDFF:FE1A:D121:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56

Pinging 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56 with 32 bytes of
data:

Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time=10ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time<1ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time=3ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time<1ms TTL=127

Ping statistics for 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Top

GeeksforGeeks

b. Use ping6 and traceroute6 to test connectivity.





c. **AIM:** Analyze ICMP error and echo messages with Wireshark

```
Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0x28de [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: fe80::1
    ICMPv6 Option (Source link-layer address : c8:53:09:63:4e:06)
        Type: Source link-layer address (1)
        Length: 1 (8 bytes)
        Link-layer address: LCFCElectron_63:4e:06 (c8:53:09:63:4e:06)
```

d. **AIM:** Compare IPv4 and IPv6 header

```
Internet Protocol Version 4, Src: 192.168.1.15, Dst: 23.212.254.56
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 1424
    Identification: 0x4c02 (19458)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.15
    Destination Address: 23.212.254.56
    [Stream index: 26]
```

```
Internet Protocol Version 6, Src: fe80::7c2b:9ebf:65dd:21be, Dst: ff02::16
    0110 .... = Version: 6
    .... 0000 0000 .... .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
        .... 0000 00.. .... .... .... .... .... = Differentiated Services Codepoint: Default (0)
        .... .... ..00 .... .... .... .... .... = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
    Payload Length: 36
    Next Header: IPv6 Hop-by-Hop Option (0)
    Hop Limit: 1
    Source Address: fe80::7c2b:9ebf:65dd:21be
        [Address Space: Link-Local Unicast]
        [Special-Purpose Allocation: Link-Local Unicast]
    Destination Address: ff02::16
        [Address Space: Multicast]
        [.... .... 0000 .... = Multicast Flags: 0x0]
            .... .... 0... .... = Reserved: 0
            .... .... .0.. .... = Rendezvous Point (RP): False
            .... .... ..0. .... = Network Prefix: False
            .... .... ...0 .... = Transient: False
        [.... .... .... 0010 = Multicast Scope: Link-Local scope (0x2)]
    [Stream index: 18]
    IPv6 Hop-by-Hop Option
        Next Header: ICMPv6 (58)
        Length: 0
        [Length: 8 bytes]
        Router Alert
            Type: Router Alert (0x05)
            Length: 2
            Router Alert: MLD (0)
        PadN
            Type: PadN (0x01)
                00.. .... = Action: Skip and continue (0)
                ..0. .... = May Change: No
                ...0 0001 = Low-Order Bits: 0x01
            Length: 0
            PadN: <none>
```

Practical 07

Practical 7: Transport Layer Protocol Testing

a. **AIM:** Use Wireshark to analyze a TCP three-way handshake.

```
Transmission Control Protocol, Src Port: 51302, Dst Port: 7680, Seq: 0, Len: 0
    Source Port: 51302
    Destination Port: 7680
    [Stream index: 3]
    [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete, SYN_SENT (1)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 526380124
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 (SYN)
    Window: 65535
    [Calculated window size: 65535]
    Checksum: 0xf8ea [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permit
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 0]
    [Server Contiguous Streams: 1]
```

b. **AIM:** Numerical on UDP and TCP header

UDP dump Structure

5M Q. XX XX XX XX XX XX XX XX

Source port | Destination port | length | checksum

given a dump of UDP header in hexa format

06 32 00 0D 00 1C E2 17 find the following

a) Source code number
b) Destination — " —
c) Length of user datagram
d) Length of data.
e) UDP header diagram

Solution:-

UDP dump structure -

XX XX XX XX XX XX XX XX

Source port | Destination port | length | checksum

a) Source code -

It is port that identify number from sender side. It helps reciever to understand which application is sending the data.

It is a 2 ~~bit~~ byte (16 bits) long field.

for the given dump the source code

$2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

16 8 4 2 1

0 1 1 0

is equal to 06 32

$(06\ 32)_{16} =$

1024
512
32
16
2
1586

$(06\ 32)_{16} = \cancel{(812)_{10}}\ (1586)_{10}$

0000 0110 0011 0010

0110
0011
0010

B = 1-0
D = 1-9

10 - A
11 - B
12 - C
13 - D

(2) Destination port :-

$(00\ 0D)_{10} = 13$

(3) Length :-

28

Size Header = 8 bytes
Total length = 28 bytes
length of data = Total length − header
= 28 − 8
= 20 bytes

Data travers = Client → Servers

Source (16 bits)	Destination port (16 bits)
length 16 bits	Checksum 16 bits
8 byts	Data

Q. 053200217 000000001 000000000 500207FF
000000000

TCP Header diagram :-

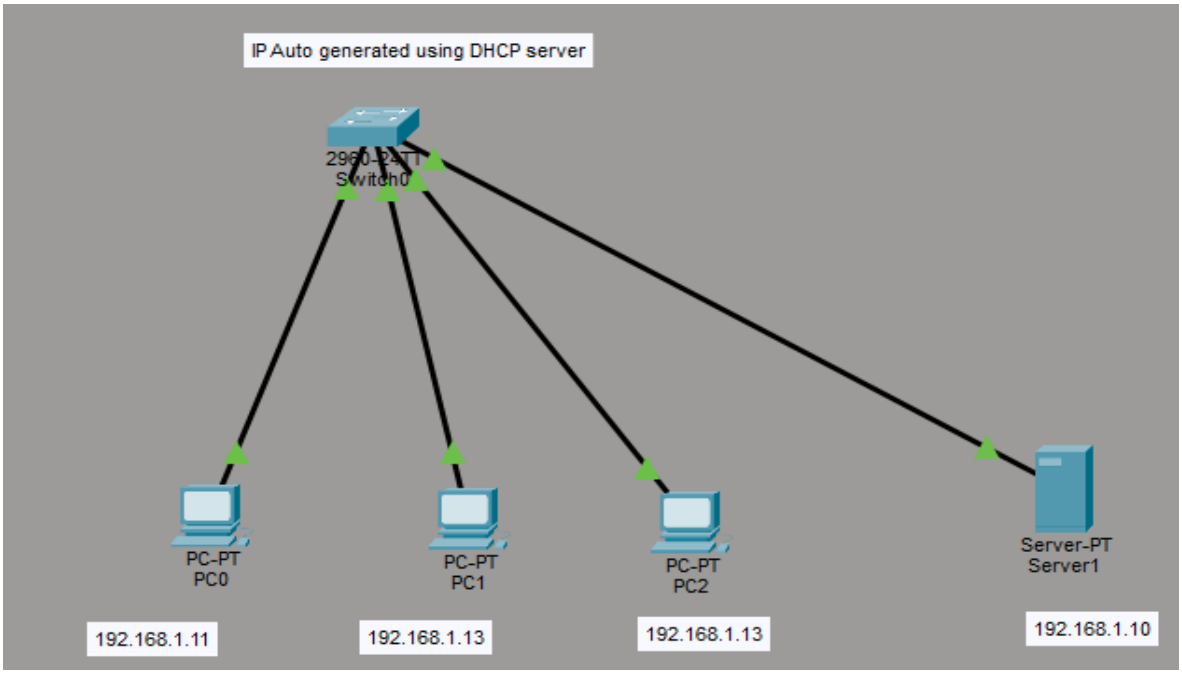
Source 16					Destination 16			
Sequence Number 32								
Acknowledgment Number 32								
HLen 4 bits	Round 6	URG	ACK	PSH	RST	SYN	FIN	windowsize 16
Checksum 16								Urgent point 16
optional Padding (40 bytes)								

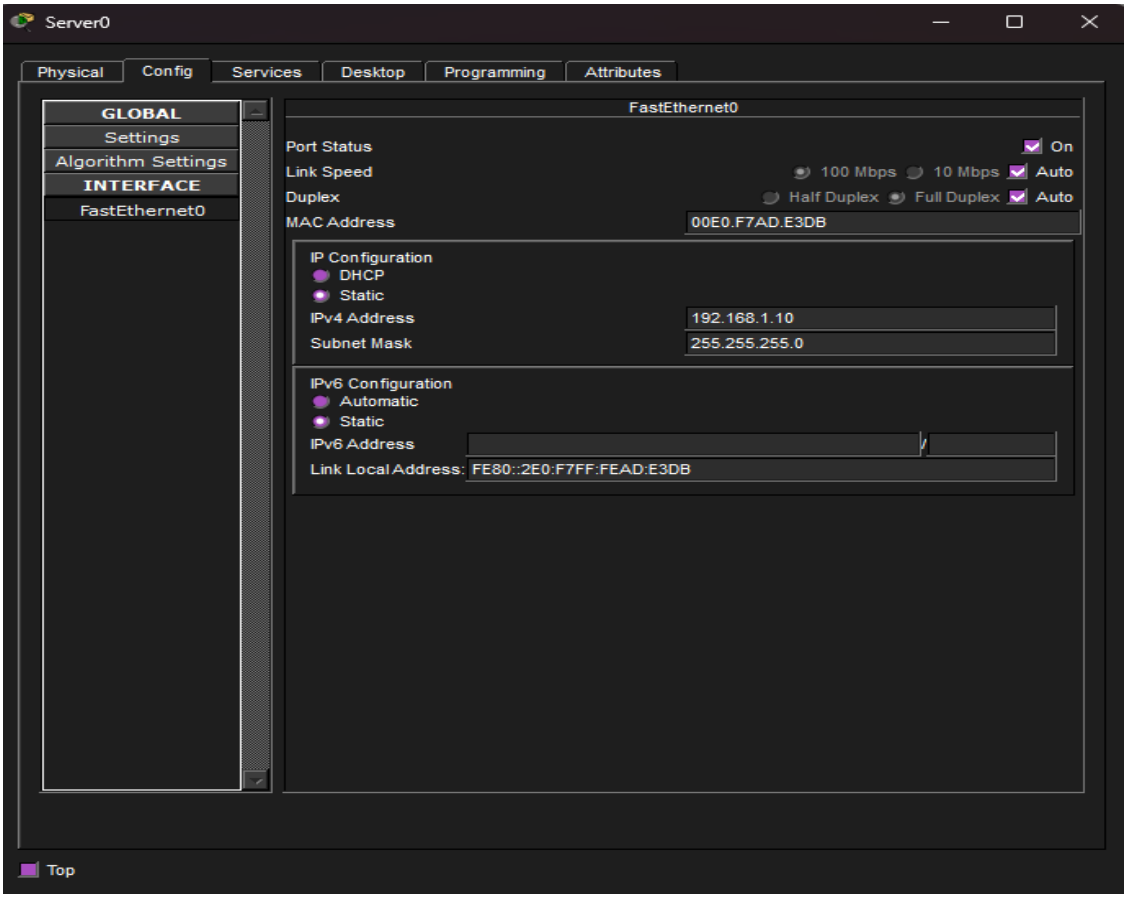
TCP dump header structure.

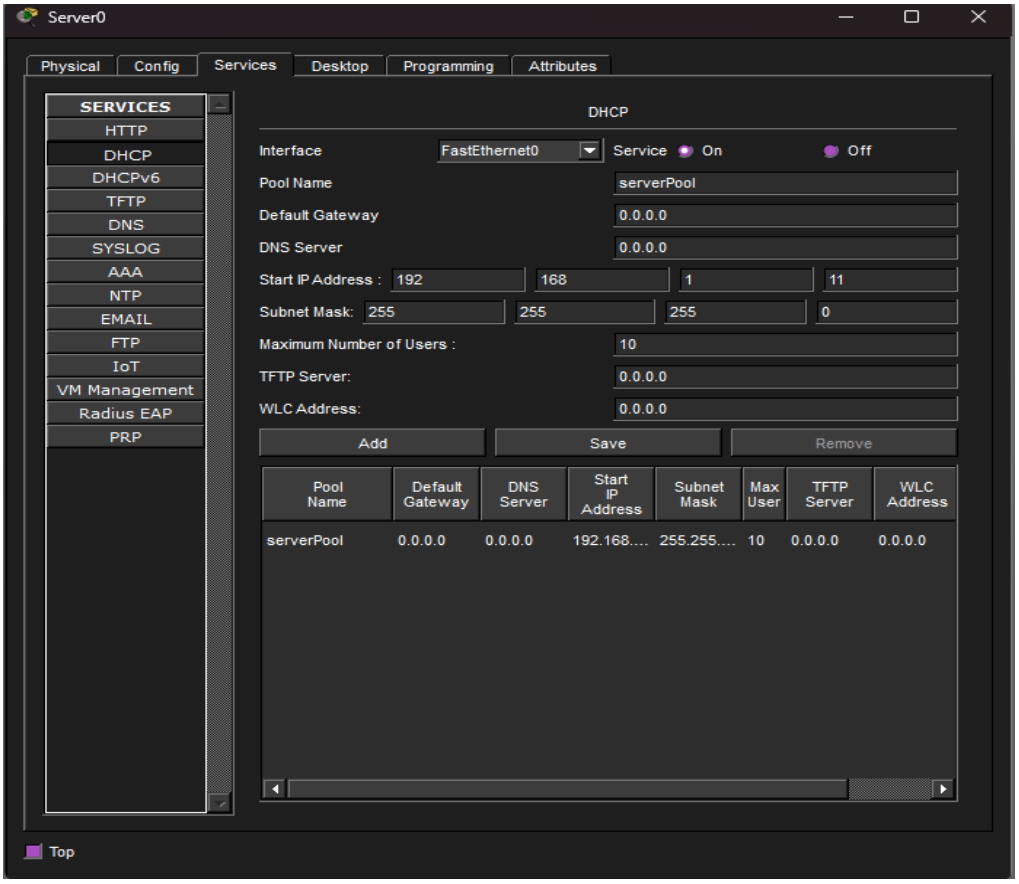
Practical 08

Practical 8: DHCP and IP Allocation Technique

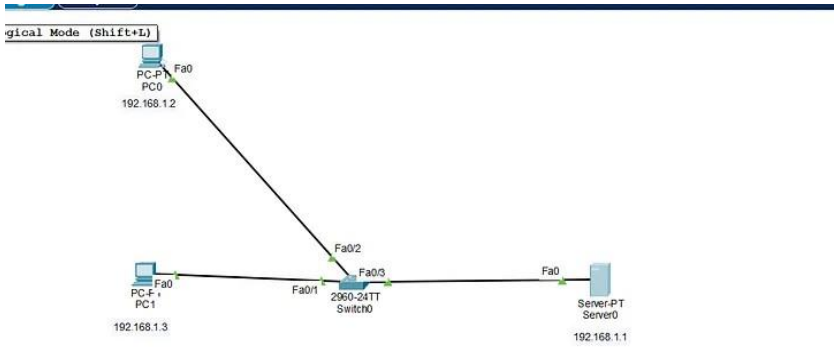
a. **AIM: Configure a DHCP server in Cisco Packet Tracer.s**

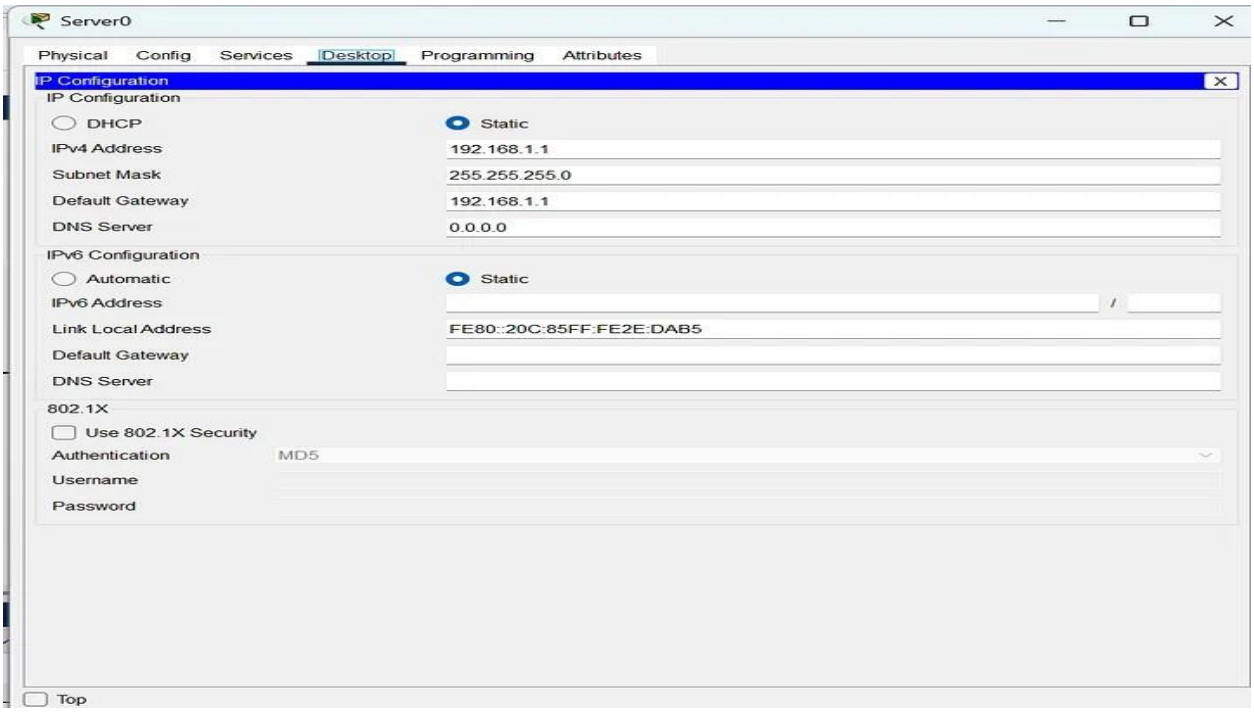



Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F7AD.E3DB
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2E0:F7FF:FEAD:E3DB
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
192
168
1
11
Subnet Mask:
255
255
255
0
Maximum Number of Users :
10
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
192.168....
255.255....
10
0.0.0.0
0.0.0.0
Top

b. **AIM:** Assign IP addresses dynamically to multiple clients and verify using ipconfig.





PC0

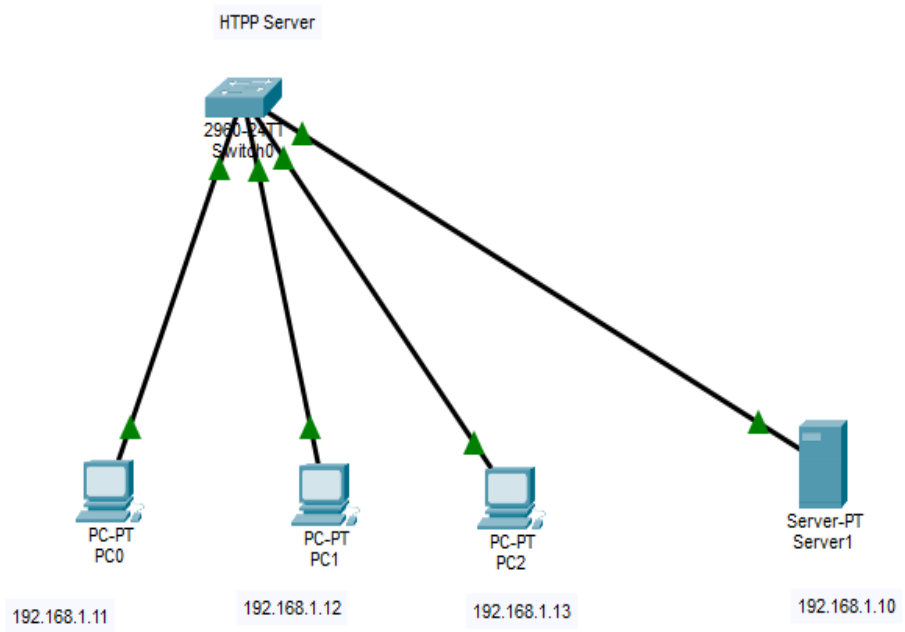
Physical Config Desktop Programming Attributes

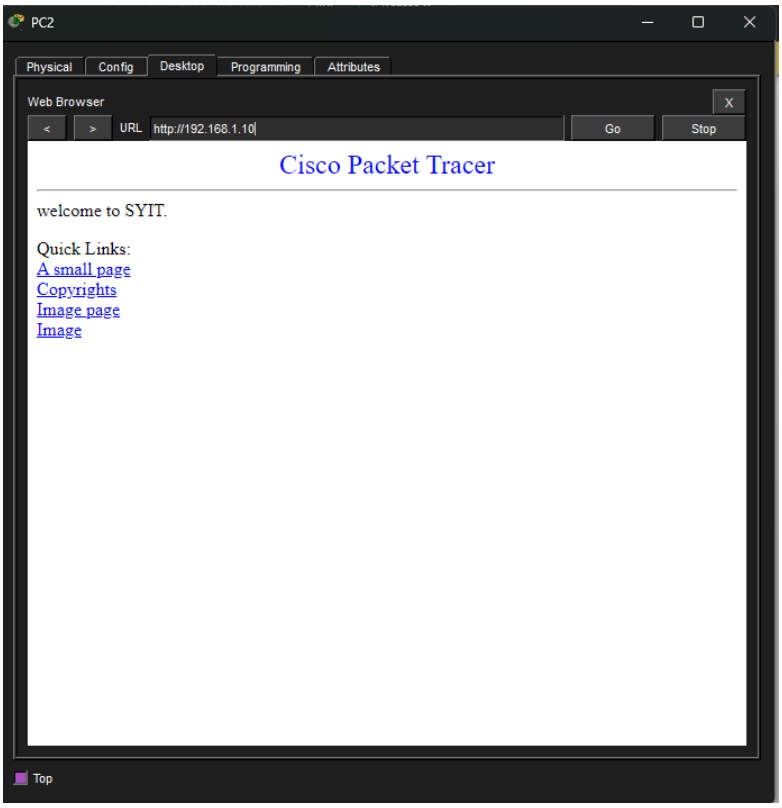
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

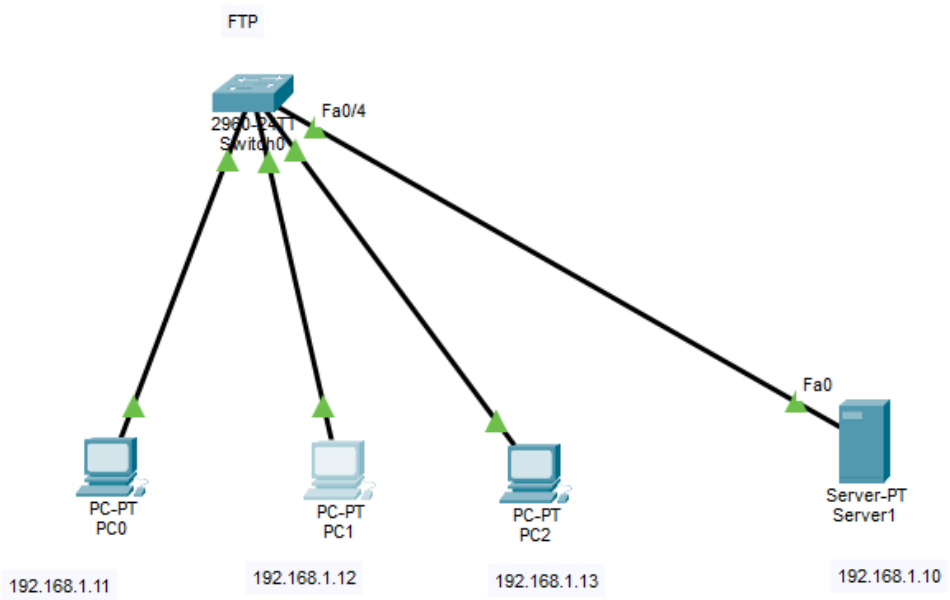
Practical 9: Application Layer Protocols

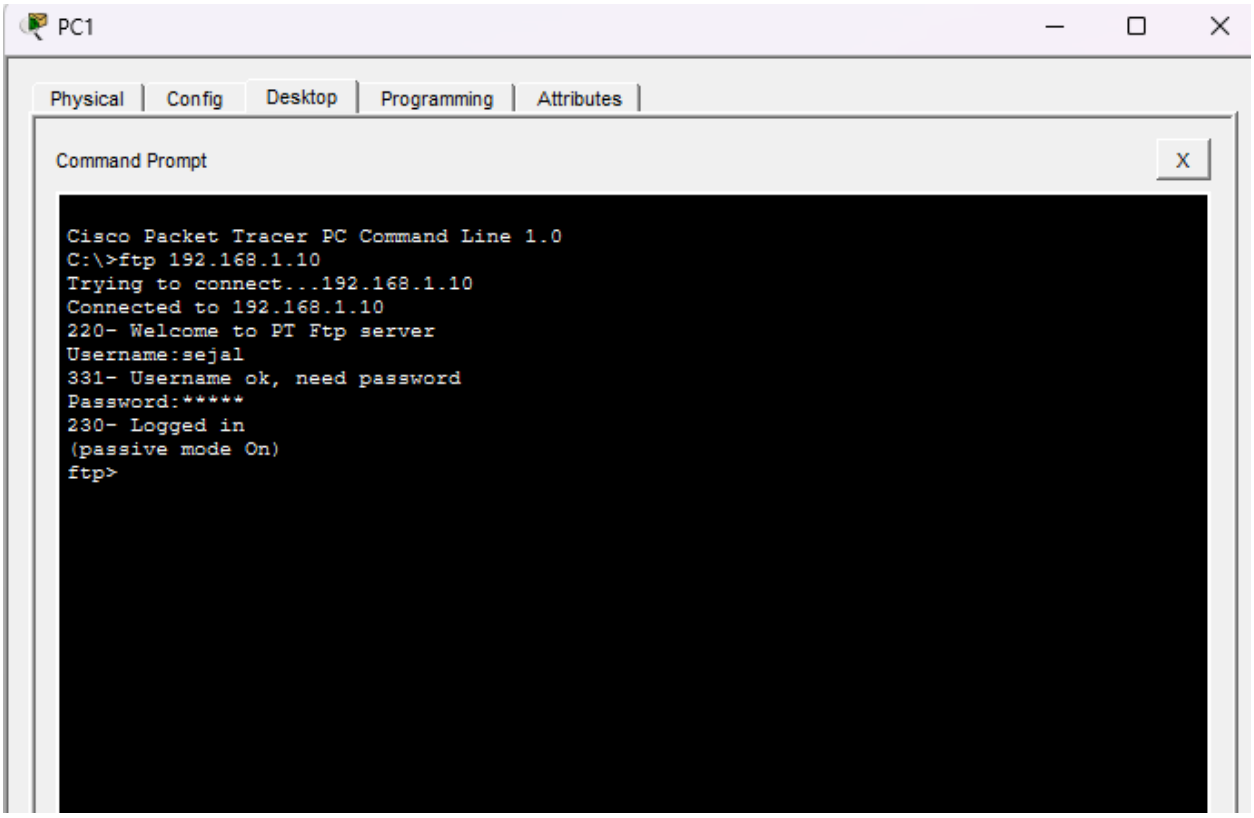
a. AIM: Set up DNS SeS Server





b. AIM: . Set up an FTP server and download/upload files.





PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
ftp>quit

221- Service closing control connection.
C:\>ftp 192.168.1.10
Trying to connect...192.168.1.10
Connected to 192.168.1.10
220- Welcome to PT Ftp server
Username:sejal
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.10:
0    : asa842-k8.bin                                    5571584
1    : asa923-k8.bin                                    30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3    : c1841-ipbase-mz.123-14.T7.bin                    13832032
4    : c1841-ipbasek9-mz.124-12.bin                     16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7    : c2600-i-mz.122-28.bin                            5571584
8    : c2600-ipbasek9-mz.124-8.bin                      13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11   : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12   : c2800nm-ipbasek9-mz.124-8.bin                    15522644
13   : c2900-universalk9-mz.SPA.155-3.M4a.bin           33591768
14   : c2950-i6q412-mz.121-22.EA4.bin                   3058048
15   : c2950-i6q412-mz.121-22.EA8.bin                   3117390
16   : c2960-lanbase-mz.122-25.FX.bin                   4414921
17   : c2960-lanbase-mz.122-25.SEE1.bin                 4670455
18   : c2960-lanbasek9-mz.150-2.SE4.bin                 4670455
19   : c3560-advipservicesk9-mz.122-37.SE1.bin          8662192
20   : c3560-advipservicesk9-mz.122-46.SE.bin           10713279
21   : c800-universalk9-mz.SPA.152-4.M4.bin             33591768
22   : c800-universalk9-mz.SPA.154-3.M6a.bin            83029236
23   : cat3k_caa-universalk9.16.03.02.SPA.bin           505532849
24   : cgr1000-universalk9-mz.SPA.154-2.CG              159487552
25   : cgr1000-universalk9-mz.SPA.156-3.CG              184530138
26   : ie9k_iosxe.17.09.04.SPA.bin                      596133776
```

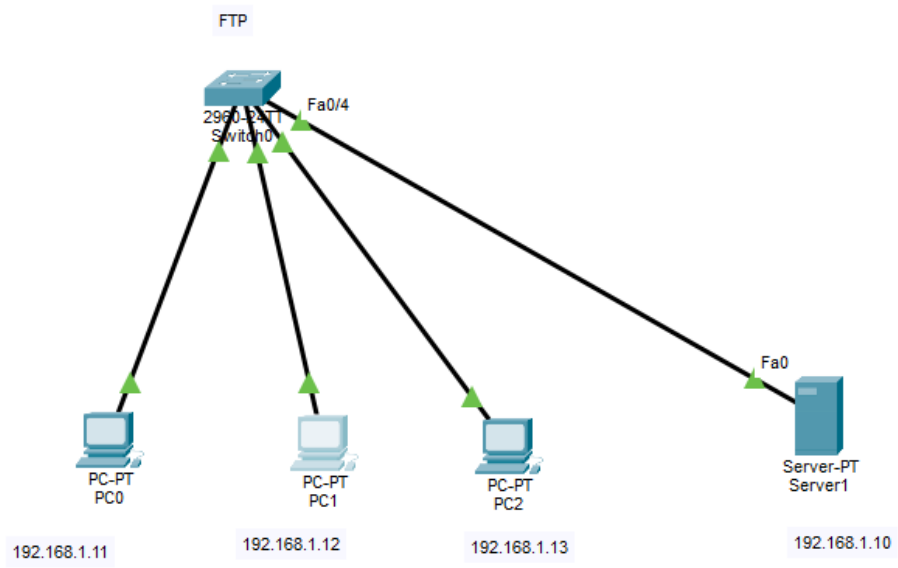
Top

c. **AIM:** Configure a basic SMTP and POP3 email exchange (simulation).

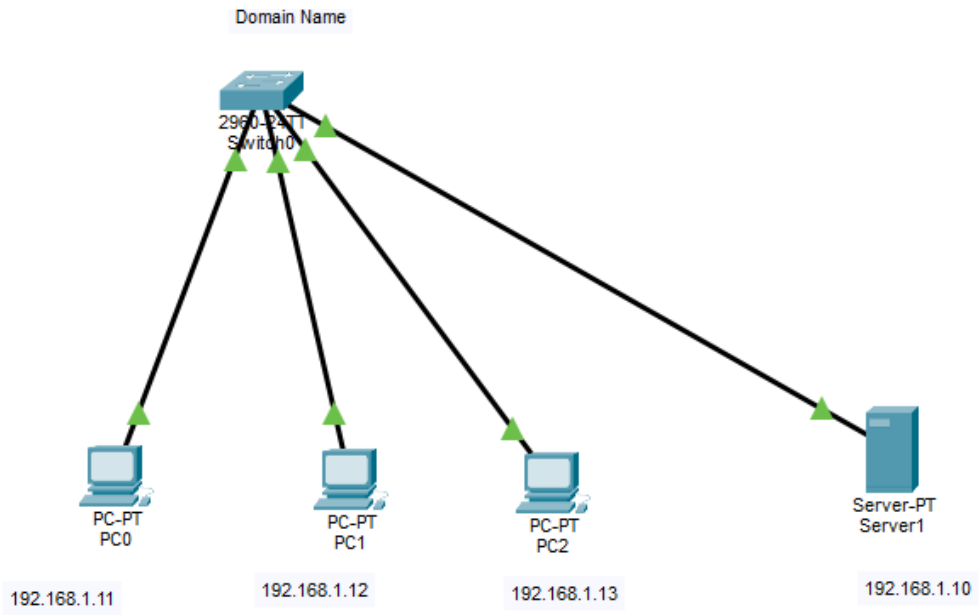
IOS Command Line Interface

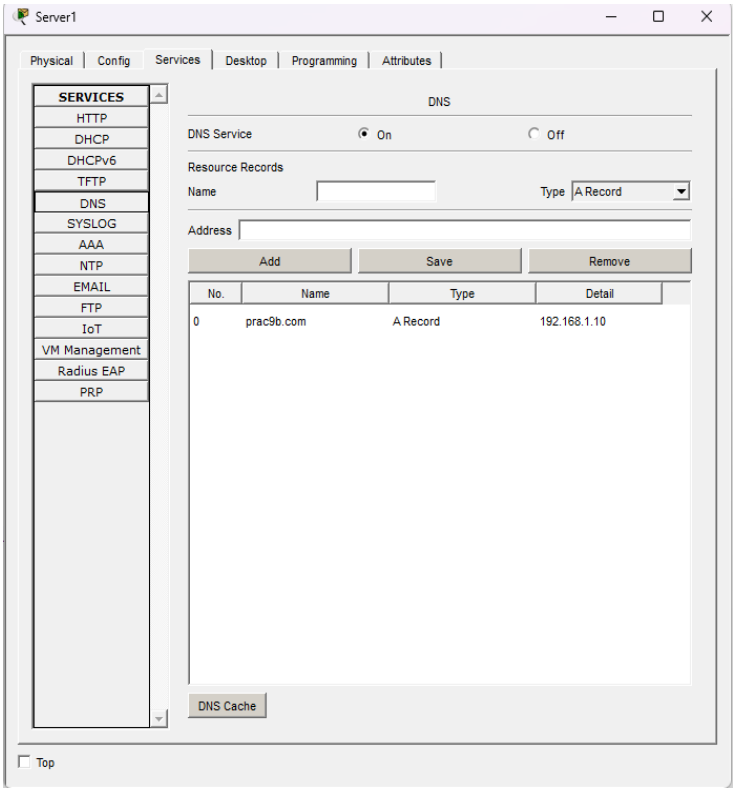
```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed sta
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#int
Router(config)#interface gi
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip add 192.168.1.100 255.255.255.0
Router(config-if)#no shut
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed sta
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

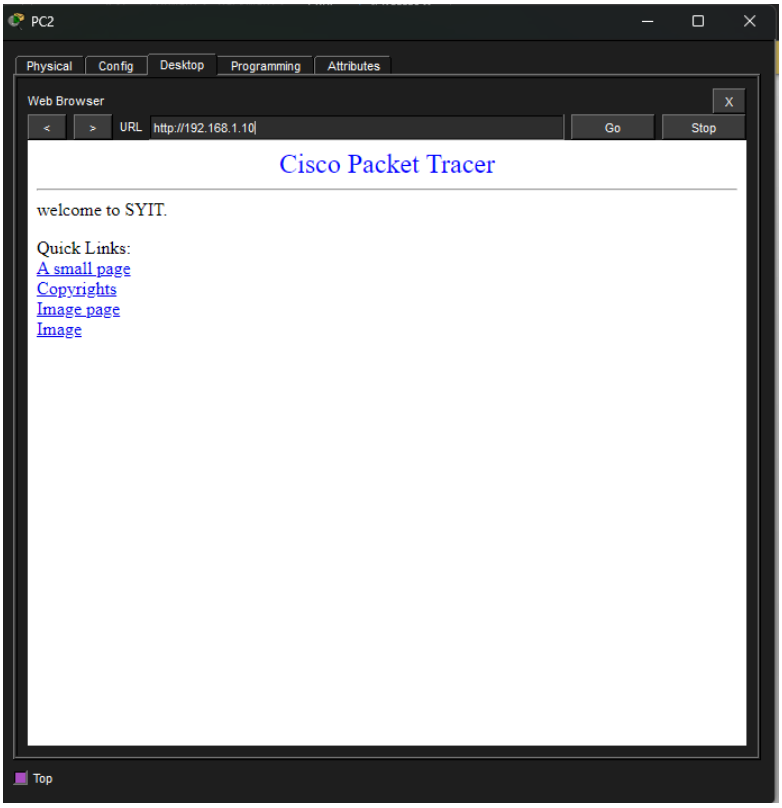
Copy



d. **AIM:** Access a website and inspect HTTP headers using browser tools.

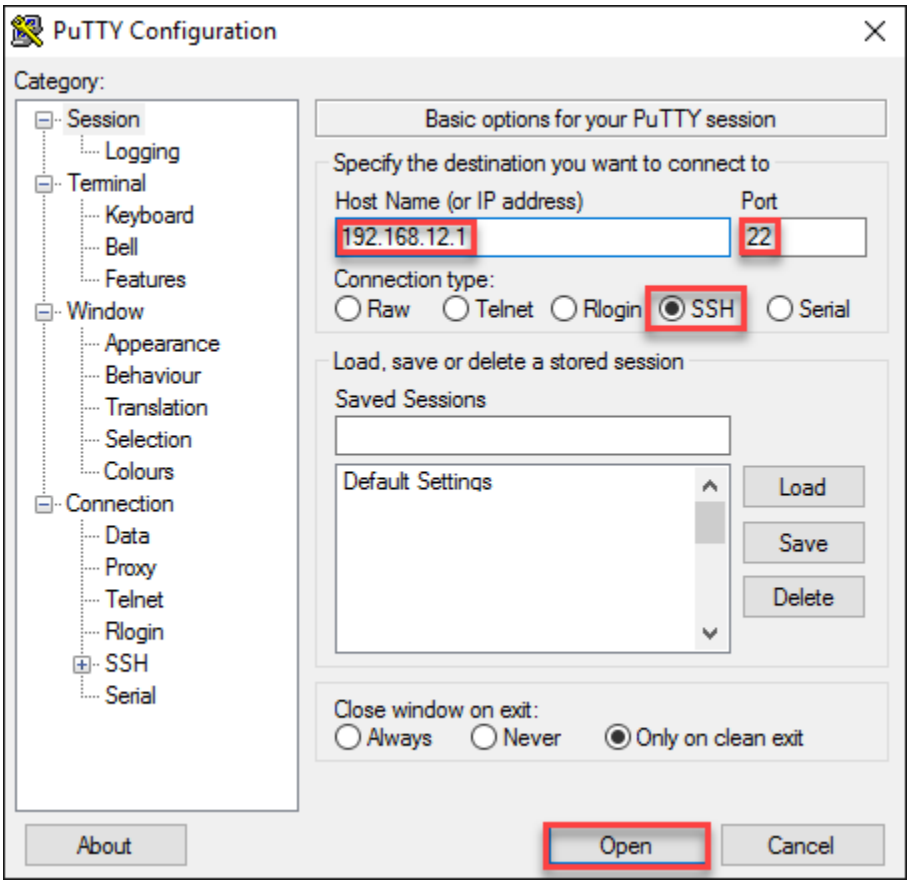





PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.1.10
Go
Stop
Cisco Packet Tracer
welcome to SYIT.
Quick Links:
A small page
Copyrights
Image page
Image
Top

Practical 10: Remote Access and Secure Connections

a. **AIM:** Connect to a remote server using TELNET (simulation).

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet
Cisco Packet Tracer PC Telnet

Usage: telnet target [port]

C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open


User Access Verification

Username: cisco
Password:
% Password:  timeout expired!
% Login invalid

[Connection to 192.168.1.1 closed by foreign host]
C:\>
```

b. Establish an SSH connection with authentication.